COPPERSTONE UNIVERSITY
School of Physical Sciences and Mathematics
Department of Electrical Engineering

Simulation-Based Design and Evaluation of a Machine Learning-Assisted Self-Healing Protection Scheme for a 33/11 kV Mining Distribution Feeder

A Thesis submitted in partial fulfilment of the requirements for the Degree of
Bachelor of Engineering in Electrical Engineering

By
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2026

Declaration of Originality

I, Victoire Chinyanta Chimundu, student number CU-BEE-100-7229, hereby declare that this thesis, submitted in partial fulfilment of the requirements for the degree of Bachelor of Engineering in Electrical Engineering at Copperstone University, School of Physical Sciences and Mathematics, is my own original work.

- This work has not been submitted previously for any degree or examination at this or any other university.
- All sources used or quoted have been acknowledged by means of complete references, and all assistance received has been disclosed.
- The simulation models, datasets, and machine learning scripts described in this thesis were developed solely by the author using MATLAB R2024a, except where explicitly acknowledged.
- Where editorial or language assistance has been used, responsibility for the technical content, interpretation of results, conclusions, and final submission remains solely with the author.

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Approval Page

This thesis by Victoire Chinyanta Chimundu, student number CU-BEE-100-7229, entitled "Simulation-Based Design and Evaluation of a Machine Learning-Assisted Self-Healing Protection Scheme for a 33/11 kV Mining Distribution Feeder", is submitted in partial fulfilment of the requirements for the Bachelor of Engineering in Electrical Engineering at Copperstone University.

The work has been reviewed and is approved for submission to the Department of Electrical Engineering.

Supervisor: ______________________________ Date: _________________

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Dean of School: ___________________________ Date: _________________

Dedication

To my father, Alexandre Kelelema Tshakamisha:
the man whose work first showed me what electrical engineering looks like in practice.
Watching you carry out your duties as an instrumentist at Mutanda Mining
planted the seed for everything in this thesis.
Your dedication to your craft was my first and most lasting lesson.

— V.C.C.

Acknowledgements

First and above all, I give thanks to God, from whom all understanding comes. This work would not have been possible without the ability, perseverance, and clarity of mind that I believe are gifts beyond my own making.

I owe a sincere debt of gratitude to my supervisor, Mr. Charles Kasonde, for his guidance, patience, and technical insight throughout the research process. His direction kept this work grounded and focused, and his willingness to engage with both the simulation and the machine learning components was invaluable.

To my mother, thank you for the emotional support you have given me throughout my studies. Your steadiness and encouragement on the difficult days carried more weight than you know.

To my father, beyond the dedication written elsewhere in this thesis, I want to acknowledge your financial support across all these years of study. Without that foundation, none of this would have been possible.

To my brother and sister, thank you for the advice and motivation you offered through every stage of this journey. Having people who know you well enough to tell you the truth, and who push you forward, nonetheless, is something I do not take for granted.

To my friend and brother, Ruben Kalunga, thank you for your emotional support throughout this journey. Even when you were not physically present, your encouragement still reached me and mattered deeply. Our gaming sessions became a remedy during stressful moments and helped me find balance when the pressure of this work felt heavy.

To Celine Muganza, thank you for the encouragement and energy you brought at every turn. Your belief in me, even when mine wavered, meant a great deal.

To Mr. Gad Naweji, thank you for your support and the role you played along the way. Your contribution to my development during this period is genuinely appreciated.

To all my friends, both near and far, thank you for your companionship, encouragement, understanding, and the moments of laughter that made this journey lighter. Your support, whether through a thoughtful word, a welcome distraction, or simply being present, helped me persevere, and I am sincerely grateful to each of you.

Finally, I acknowledge Copperstone University and the School of Physical Sciences and Mathematics for providing the academic environment and resources that made this research possible.

Victoire Chinyanta Chimundu
Copperstone University, 2026

Abstract

Mining distribution feeders supply safety-critical and production-critical loads, including dewatering pumps, ventilation systems, crushers, conveyors, and mineral-processing plant. In radial 33/11 kV mining networks, conventional time-overcurrent protection clears faults reliably but provides neither bus-level fault localisation nor automated service restoration. Consequently, healthy downstream loads can remain de-energised until the faulted section is manually located, isolated, and returned to service.

This thesis presents a simulation-based, proof-of-concept design and evaluation of a machine learning-assisted fault isolation and service restoration scheme for a representative five-bus 33/11 kV mining distribution feeder. The feeder, its two 33/11 kV transformers, four industrial load centres, circuit breakers, fault blocks, RMS measurement points, and a normally-open tie-switch were modelled in MATLAB R2024a Simulink. A simulation-based approach was adopted because controlled fault testing on live high-voltage mining feeders is unsafe, costly, and inaccessible within the available undergraduate infrastructure.

A synthetic dataset of 1,000 labelled samples was generated from controlled simulations of healthy operation and of single line-to-ground, line-to-line, and three-phase faults at four bus locations. Each sample is described by a 24-element feature vector of three-phase RMS voltages and currents measured at the monitored buses. A cost-sensitive Random Forest classifier was trained to identify fault type and location, and its prediction was mapped to selective circuit-breaker isolation and, where the topology permits, tie-switch restoration. Every switching decision in the evaluation was driven by the classifier prediction computed from the pre-switch measurements, forming a closed detection-to-restoration loop rather than a look-up against the known fault.

Under the controlled near-bolted fault conditions studied, the classifier achieved a zero-error result on the held-out test set; the predicted fault zone was correct in all twelve restoration scenarios; the faulted zone was isolated in every case; the tie-switch was never closed into a terminal-zone (B4 or B5) fault; and the six restoration-eligible cases held post-restoration voltage within the 0.95-1.05 pu band (0.963-0.981 pu). These outcomes establish feasibility within the simulated parameter space only; they must not be interpreted as field-validated performance, as the study relies on synthetic data, ideal sensors, a discrete parameter grid, and does not represent NER-limited earth faults. Hardware-in-the-loop testing, NER-aware feature design, and validation against field data are recommended before any practical deployment is considered.

Keywords: machine learning, Random Forest, self-healing protection, fault classification, mining distribution feeder, selective isolation, MATLAB/Simulink, Zambian Copperbelt.

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Nomenclature

A. Abbreviations

3PH	Three-phase balanced fault
CB	Circuit breaker
CT / VT	Current transformer / voltage transformer
CTI	Coordination time interval
CV	Cross-validation
FDIR	Fault detection, isolation and restoration
F1	F1-score (harmonic mean of precision and recall)
HIL	Hardware-in-the-loop
IDMT	Inverse definite minimum time
IED	Intelligent electronic device
LL	Line-to-line fault
ML	Machine learning
NER	Neutral earthing resistor
N/O	Normally open
OOB	Out-of-bag
pu	Per unit
RF	Random Forest
RMS	Root mean square
RTDS	Real-time digital simulator
SLG	Single line-to-ground fault
SX-EW	Solvent extraction and electrowinning
TOC	Time-overcurrent (protection)

B. Symbols

$\mathbf{V_{pu}}$	Per-unit RMS voltage at a monitored bus
$\mathbf{V_{base}}$	Line-to-line RMS base voltage (11 kV on the feeder side)
$\mathbf{S}$	Three-phase apparent power, $S = \sqrt{3} \cdot V_{LL} \cdot I_L$
$\mathbf{S_{restored}}$	Apparent power reconnected through the tie-switch
$\mathbf{S_{T2,rated}}$	Rated apparent power of transformer T2 (15 MVA)
$\boldsymbol{\lambda_{T2}}$	T2 loading ratio, $S_{restored} / S_{T2,rated}$
$\mathbf{x}$	24-element RMS feature vector (voltages and currents at B2-B5)
$\mathbf{f_{RF}}$	Trained Random Forest classification function
$\hat{\mathbf{y}}$	Predicted class label (0 = Healthy; 1-12 = fault classes)
$\mathbf{M}$	Number of trees in the forest (M = 500)
$\mathbf{u_{tie}}$	Tie-switch command (1 = close, 0 = remain open)
$\mathbf{N_{healthy,off}}$	Healthy zones remaining de-energised after switching
$\mathbf{R_V}$	Voltage recovery ratio, $V_{post} / V_{pre\text{-}fault}$

Chapter 1 Introduction

1.1 Background and Context

1.1.1 Electrical Power in Mining Operations

Modern mining operations are among the most electrically intensive industrial activities in the world. Unlike many manufacturing processes, mining relies on uninterrupted electrical supply because the consequences of a power cut extend well beyond lost production. Underground dewatering systems must run continuously to keep active workings safe from groundwater flooding. Ventilation fans must supply breathable air to underground workers without pause. Crushers, conveyors, hoisting equipment, and mineral-processing plant cannot be stopped abruptly without causing mechanical damage or losing material mid-process. In a large Copperbelt copper mine, the cost of a single hour without power can run to tens of thousands of dollars in lost production alone, before accounting for the safety consequences of failed ventilation or dewatering [1].

The electrical infrastructure serving these loads in the Zambian Copperbelt is built around medium-voltage distribution feeders operating at 33/11 kV. These feeders extend across mine sites, sometimes over several kilometres, connecting main substation transformers to individual load centres drawing between one and three megawatts each. Most feeders are radial in topology, a straightforward, cost-effective arrangement that suits the geographic spread of mining loads but carries a significant protection limitation that this research sets out to address [2].

1.1.2 The Harsh Operating Environment

Mining feeders face considerably more demanding operating conditions than urban distribution networks. Long underground cable runs are exposed to mechanical damage from mining equipment, moisture, dust, vibration, and elevated ambient temperatures. Surface feeders contend with lightning strikes, wildlife, and the physical impacts of blasting activity. The combined effect of these conditions is a fault rate measurably higher than urban equivalents, and a production environment in which every fault event carries serious consequences [3].

The dominant fault modes in mining systems are single line-to-ground (SLG) faults caused by insulation breakdown or cable damage, line-to-line (LL) faults from conductor contact, and three-phase (3PH) faults from direct short-circuit events such as cable strikes. SLG faults are the most common, accounting for the majority of fault incidents in medium-voltage industrial networks worldwide [4].

1.1.3 Limitations of Conventional Protection

The protection technology that has served mining feeders in the Copperbelt for decades is the time-overcurrent relay operating to the Inverse Definite Minimum Time (IDMT) characteristic. In a radial feeder, a cascade of such relays is coordinated through time-grading: the relay nearest the fault operates first, and upstream relays are held back by progressively longer time delays. This scheme is reliable, well understood, and inexpensive [5].

Its decisive limitation is the absence of spatial discrimination: the scheme clears a fault but cannot identify the bus at which it occurred. The responding relay therefore de-energises the entire downstream section, including electrically healthy buses whose loads could have remained in service. Supply is restored only after a maintenance crew has located the fault, isolated the affected section, and manually re-closed the healthy sections, a sequence that routinely consumes 15 to 60 minutes, during which every downstream load is interrupted irrespective of its involvement in the fault [6].

1.2 Motivation for the Research

A materially better outcome is possible if protection can perform two functions that conventional overcurrent schemes cannot: identify the specific bus section that has faulted, and use that information to isolate only the faulted section while keeping the healthy sections energised. Where a healthy bus is de-energised as a side effect of isolation, supply can be further restored by closing a normally-open tie-switch onto an alternative path, reducing the effective outage of healthy loads from minutes to seconds. This capability is topology-bounded: in the five-bus feeder studied, restoration of the downstream terminal bus B4 is achievable for upstream faults at B2 or B3, while the intermediate faulted bus itself remains de-energised until manual intervention. This partial-restoration behaviour is a structural property of single-tie-switch radial feeders, examined further in Chapters 3 and 6, and represents the Fault Detection, Isolation, and Restoration (FDIR) framework of Srivastava et al. [2] adapted to that topological constraint.

Fault localisation of this kind is a pattern-recognition problem to which machine learning is well suited. A classifier trained on simulated healthy and faulted operating conditions can learn the mapping between the three-phase RMS signature measured across several buses and the type and location of the fault that produced it. Once trained, inference is effectively instantaneous, allowing the location decision to be produced on a timescale far shorter than manual diagnosis [8].

1.3 Research Objectives

The main objective of this research is to design and evaluate, through MATLAB/Simulink simulation, a machine learning-assisted fault isolation and service restoration scheme for a representative 33/11 kV mining distribution feeder. The specific objectives are:

1. To model a representative five-bus 33/11 kV mining distribution feeder in MATLAB R2024a Simulink.
2. To generate simulated fault data for selected SLG, LL, and three-phase faults under controlled operating conditions.
3. To train and evaluate a Random Forest classifier using RMS voltage and current features for fault-type and fault-location identification.
4. To assess whether the proposed isolation and tie-switch restoration logic can disconnect the faulted section and restore eligible healthy loads within acceptable voltage limits.

1.4 Research Questions

1. Can RMS voltage and current features from multiple buses be used to classify the simulated feeder state and identify the faulted section under controlled near-bolted fault conditions?
2. Does cost-sensitive Random Forest classification reduce the risk of missed simulated faults while maintaining acceptable overall classification performance?

3. Can the proposed switching logic restore eligible healthy load sections through the normally-open tie-switch while maintaining post-restoration voltage within the 0.95-1.05 pu band?

1.5 Scope and Delimitations

This study is a proof-of-concept simulation-based evaluation. It is conducted in MATLAB R2024a Simulink using the Simscape Electrical Specialized Power Systems library. The reported results were produced with MATLAB R2024a (version 24.1.0.2537033) - Simulink, Simscape Electrical and the Statistics and Machine Learning Toolbox - on a Windows 11 workstation with an Intel Core i5-6200U processor and 8 GB of RAM. The feeder model is limited to a five-bus radial topology with one normally-open tie-switch, ideal RMS measurements, and simulated SLG, LL, and three-phase faults. The study does not include field measurements, physical relay testing, hardware-in-the-loop validation, or mine-site implementation.

Fault simulation is limited mainly to near-bolted and moderate-impedance conditions. The model does not fully represent the NER-limited 1-5 A SLG fault regime used in many properly earthed Copperbelt 11 kV installations. The classification results should therefore not be generalised directly to NER-earthed feeders without additional modelling, new features such as zero-sequence and negative-sequence quantities, and retraining on a representative NER-limited dataset.

1.6 Thesis Structure

Chapter 2 reviews published literature on mining feeder protection, self-healing distribution automation, and machine learning for fault classification. Chapter 3 describes the research methodology, including the research design and its justification, the mathematical formulation, the feeder model specification, the dataset generation strategy, the classifier design, the isolation and restoration logic, and the ethical, safety and data-integrity considerations. Chapter 4 documents the MATLAB/Simulink implementation, including the measurement and grounding configuration verified during validation. Chapter 5 presents and analyses the simulation results across electrical behaviour, ML classifier performance, self-healing restoration, and a comparative benchmark against conventional protection. Chapter 6 states conclusions, assesses objective achievement, answers the research questions, acknowledges limitations, and recommends future work. Supporting reproducibility material and key parameter summaries are provided in the appendices.

Chapter 2 Literature Review

2.1 Introduction

This chapter reviews literature relevant to three linked areas: medium-voltage mining feeder protection, self-healing distribution automation, and machine-learning-based fault classification. The review is not limited to describing previous studies; it evaluates how far those studies solve the engineering problem addressed in this thesis. Particular attention is paid to selectivity, restoration capability, data realism, earthing practice, and the gap between high simulated classification accuracy and practical protection deployment. This critical approach is necessary because a protection scheme for a mining feeder cannot be judged only by whether it detects a fault; it must also isolate the correct section, avoid unnecessary loss of supply to healthy process loads, and remain credible under the infrastructure limitations of the Zambian Copperbelt context.

2.2 Mining Feeder Protection

2.2.1 Time-Overcurrent Protection and its Limitations

Relays operating to the Inverse Definite Minimum Time (IDMT) characteristic remain the dominant protection on radial industrial feeders because they are simple, inexpensive, rugged, and familiar to protection engineers, and because they require neither communication infrastructure nor significant computation [1], [4]. That same simplicity is the source of their principal weakness. Coordination rests on current magnitude and time grading alone and carries no mechanism for identifying the faulted bus or section; the scheme is consequently effective for fault clearing but offers no basis for automated service restoration.

Bhattacharya et al. [4] show that relay coordination can be affected by variable fault levels and network reconfiguration. This finding is important because mining feeders are not static systems: load levels, feeder configurations, transformer loading, and operating conditions can change during production. Wen et al. [5] also highlight the long restoration delay associated with manual intervention after wide-area tripping. The limitation exposed by these studies is not merely slow operation of relays, but the absence of section-level intelligence after fault clearing. In a mining environment, a technically acceptable protection scheme should not only interrupt the fault; it should also help determine which healthy loads can remain energised or be restored safely. This limitation motivates the selective isolation and restoration focus of the present thesis.

2.2.2 High-Impedance and NER-Limited Fault Detection

High-impedance and NER-limited earth faults expose a deeper weakness in current-magnitude-based protection. In Neutral Earthing Resistor (NER) grounded systems, commonly associated with safer medium-voltage industrial practice [21], single line-to-ground fault current can be intentionally limited to low values. Under such conditions, the fault may not produce the large current rise assumed by conventional overcurrent relays or by many ML fault-classification studies. Guo et al. [9] therefore argue for methods that use the complete voltage and current signature rather than a simple threshold, an emphasis echoed by recent high-impedance fault-detection work for resonant-grounded systems [22]. The implication for this thesis is critical: an ML classifier trained mainly on near-bolted faults may perform well in simulation but cannot automatically be assumed valid for NER-limited Copperbelt feeders. This is why the present work is framed as a proof-of-concept under near-bolted fault assumptions, with NER-aware modelling treated as a necessary extension rather than a completed contribution.

2.2.3 Adaptive and Centralised Protection

Adaptive and centralised protection approaches attempt to overcome the rigidity of fixed IDMT settings by using wider measurement coverage and updated decision logic [7]. Their advantage is clear: measurements from several buses can support better fault localisation than a single relay operating independently. However, the literature sometimes underestimates the practical requirements of such schemes. Centralised protection depends on reliable measurements, time synchronisation, communication channels [20], cybersecurity, switchgear status feedback, and engineering staff capable of maintaining the system. In a resource-constrained mining feeder, these requirements cannot simply be assumed. For this reason, the architecture used in this thesis is deliberately simplified: it uses multi-bus RMS measurements and central decision logic in simulation, but it does not claim to replace a certified relay protection scheme without further hardware and field validation.

2.3 Self-Healing Distribution Network Automation

2.3.1 The FDIR Framework

Fault Detection, Isolation, and Restoration (FDIR) provides the conceptual basis for self-healing networks. Srivastava et al. [2] describe the sequence of detecting an abnormal condition, isolating the faulted section, and restoring supply to healthy sections through reconfiguration. The framework is useful because it shifts the protection objective from only clearing a fault to managing the post-fault network state. Nevertheless, most FDIR studies are developed for distribution systems with automation infrastructure, controllable switches, communication links, and reasonably well-documented feeder models. These assumptions are not always valid in older or less automated mining networks. FDIR literature therefore supports the objective of this thesis, but it also shows why a realistic undergraduate study must limit the network topology and switching actions considered.

Centralised FDIR schemes can be globally coordinated, but their effectiveness depends on the accuracy of fault localisation and on the availability of alternative supply paths. This dependence is decisive for the five-bus feeder studied here: a single normally-open tie-switch can restore the terminal section in selected cases, but it cannot restore every healthy load after an upstream fault, because the feeder topology itself constrains the available power paths. Any self-healing claim must therefore be topology-aware. The present research accordingly evaluates restoration only for eligible sections and does not claim full feeder self-healing across all fault locations.

2.3.2 Self-Healing Performance

Gao et al. [6] report that automated restoration can reduce outage duration from manual timescales to a few seconds, and recent reviews survey the range of self-healing restoration strategies now proposed [23]. Such results demonstrate the potential value of self-healing control, especially for continuous-process loads. However, the literature often gives less attention to the boundary conditions under which restoration is permissible: transformer spare capacity, voltage drop, switch ratings, fault isolation confirmation, and protection interlocking. In mining feeders, closing a tie-switch without checking these constraints could transfer overloads or energise an unsafe section. This thesis therefore uses voltage and capacity checks before restoration, treating FDIR as an engineering decision problem rather than a purely algorithmic reconfiguration task.

2.4 Machine Learning for Fault Classification

2.4.1 Feature Representation

Feature selection is one of the most important methodological choices in ML-based protection. RMS voltage and current magnitudes are attractive because they are readily available from many protective relays and measurement devices, and they avoid the computational complexity of wavelet or frequency-domain processing [10]. This makes RMS features suitable for a BEng-level simulation study and for a practical proof-of-concept. The limitation is that RMS magnitude features may hide information contained in phase angles, symmetrical components, harmonics, travelling waves, or transient patterns. A classifier based only on RMS features is consequently likely to perform best when faults cause clear voltage sag and current rise, and less reliably when earth faults are NER-limited or high impedance.

The spatial arrangement of features is also decisive. Chanda and Soltani [8] and Rizeakos et al. [14] show that measurements from multiple buses can support simultaneous fault-type and fault-location classification. This is more relevant to selective isolation than single-point detection because the classifier must distinguish where the fault occurred, not merely whether a fault exists. However, multi-bus measurement schemes introduce practical issues that many studies do not fully address sensor calibration, missing data, communication delay, inconsistent sampling windows, and the cost of installing measurement devices at several buses. The 24-element feature vector used in this thesis is therefore best understood as a controlled simulation representation of a possible measurement architecture, not as proof that the same measurement quality would be available in a mine without additional instrumentation.

2.4.2 Algorithm Comparison

Table 2.1 compares the candidate algorithms to justify the selection on technical grounds. Random Forest is adopted because it suits the data of this study: structured RMS voltage-current features, a moderate dataset size, and a requirement for interpretability during engineering assessment.

Table 2.1: Comparison of candidate machine-learning algorithms

Algorithm	Speed	Robustness	Interpretability	Data requirement
Decision tree	Fast	Low	High	Low
Support vector machine	Moderate	Moderate	Low	Moderate
Random Forest (selected)	Fast	High	Moderate	Low-moderate
Neural network (MLP)	Fast	High	Very low	High
Graph neural network	Moderate	High	Low	High

Recent comparative studies often identify Random Forest as a strong algorithm for tabular fault-classification data [11], [12], [24]. Its advantages are relevant to this thesis: it handles mixed-scale voltage and current features without strict normalisation, is less data-hungry than many neural-network approaches, provides feature-importance information, and is easier to explain than a deep-learning model. However, Table 2.1 should not be read as a universal ranking of algorithms. The best classifier depends on the feature set, network topology, fault scenarios, noise level, and validation method used.

The main weakness of Random Forest in this application is that its decision boundary is learned from the dataset rather than derived from a physical relay characteristic. Excellent simulated accuracy can therefore result from clean, repeated patterns in the training data rather than from true field robustness. A classifier trained and tested on the same structured simulation grid may learn the grid characteristics very well while still failing under unmodelled conditions such as CT saturation, noisy measurements, changing earthing practice, or unusual fault resistance. For this reason, the Random Forest model in this thesis is treated as an engineering feasibility tool, not as a certified replacement for industrial protection relays.

2.4.3 Cost-Sensitive Learning for Imbalanced Datasets

Cost-sensitive learning is attractive in protection studies because a missed fault is usually more serious than a false alarm. Rahman et al. [13] show that increasing the cost of fault-to-healthy misclassification can improve fault recall. This aligns with mining protection priorities, where failing to detect a fault may sustain unsafe conditions. However, cost weighting does not eliminate the need for engineering selectivity. A classifier that avoids missed faults but frequently trips the wrong section could still be unacceptable because it may disconnect critical healthy loads. The cost matrix in this thesis is therefore useful only when considered together with location accuracy, breaker mapping, and restoration checks.

2.4.4 Validation Requirements

Validation is necessary because a single train-test split can give a misleading impression of performance [15], [16]. Cross-validation, confusion matrices, and class-level metrics provide a more complete picture than overall accuracy alone. Nevertheless, validation cannot compensate for unrealistic data. If all samples come from an ideal simulation model with ideal sensors and a limited parameter grid, the confidence placed in the result must remain bounded by those assumptions. This distinction is important in the present thesis: the evaluation metrics measure consistency within the simulated dataset, while practical deployment would require independent validation using hardware-in-the-loop tests, relay recordings, or field data.

A recurring weakness in the reviewed ML protection literature is the limited connection between algorithmic accuracy and protection-engineering practice. Many studies emphasise high classification percentages but give less attention to earthing arrangement, relay coordination, switchgear actuation, communication reliability, fail-safe behaviour, and commissioning procedures. This creates a risk that the ML model is evaluated as a data-science problem rather than as part of a protection system. The present thesis attempts to reduce that gap by linking classification output to selective breaker isolation and tie-switch restoration, while explicitly acknowledging that the final step toward practical use would require hardware and field validation.

2.5 Research Gap

Taken together, the three bodies of work offer complementary but individually incomplete solutions. Time-overcurrent protection is robust and familiar yet provides neither fault localisation nor restoration support. FDIR supplies a sound automation framework but generally presupposes the controllable switching and communication infrastructure that resource-constrained mining feeders lack. Machine-learning classifiers recover fault patterns from multi-bus measurements, but their reported accuracy typically rests on synthetic or idealised datasets and is seldom related to field constraints such as earthing practice or sensor quality.

The gap is therefore not the novelty of applying a 13-class classifier to a five-bus feeder, but the absence of a bounded, engineering-oriented demonstration that integrates simulated medium-voltage feeder behaviour, ML-based fault-type and fault-location identification, selective isolation, and post-fault restoration assessment for a mining-style Copperbelt feeder. This thesis addresses that gap with an integrated MATLAB/Simulink proof-of-concept whose scope is deliberately confined to controlled near-bolted conditions, and which would require NER-aware features, noisy measurements, hardware-in-the-loop testing, and field-data validation before any deployment claim could be supported.

2.6 Chapter Summary

In summary, IDMT protection remains strong for fault clearing but weak for localisation and restoration; FDIR contributes the automation concept; and ML classification contributes the pattern-recognition tool. These elements are rarely integrated. This thesis adopts a bounded approach that links feeder simulation, RMS feature extraction, Random Forest classification, breaker selection, and restoration checks within a single proof-of-concept framework appropriate to the available infrastructure.

Chapter 3 Research Methodology

3.1 Research Design

The research follows a quantitative, simulation-based design in which all experimental data are produced by a MATLAB/Simulink model of the feeder. Simulation was preferred to field measurement for four reasons: faults can be applied safely and repeatably without risk to plant or personnel; the fault parameters can be controlled to give systematic coverage of the operating space; ground-truth labels are available for every sample without additional instrumentation; and the approach is compatible with an undergraduate project that has no access to live mining-feeder infrastructure. Figure 3.1 sets out the workflow followed throughout the study. The overall project schedule is presented as a Gantt chart in Appendix B.

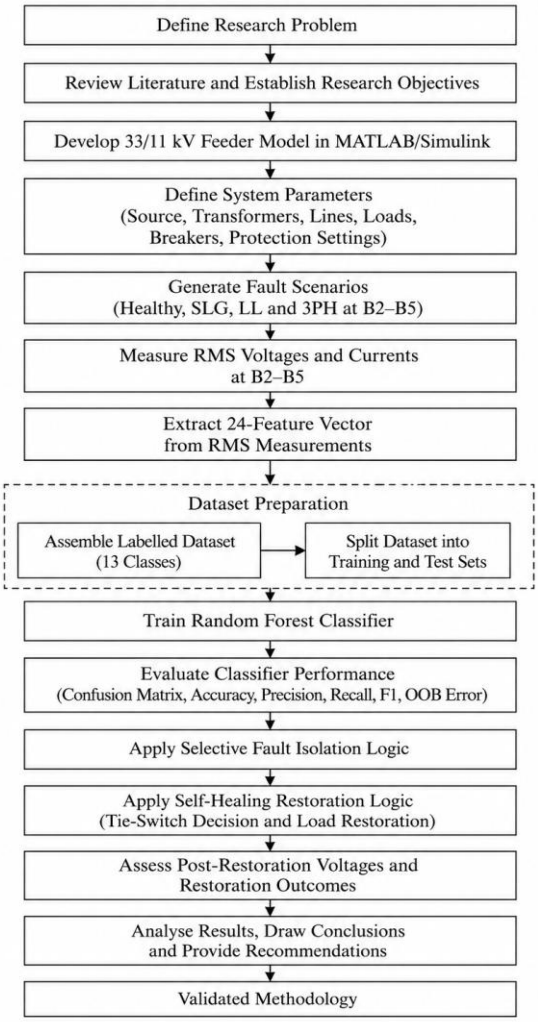

Figure 3.1: Research methodology workflow

The workflow defines the investigation as a sequence of dependent engineering stages: the feeder is modelled, faults are generated, features are extracted, the classifier is trained, and the resulting protection decision is evaluated against the simulated electrical behaviour. This ordering ensures that each result can be traced to the stage that produced it.

3.1.1 Justification for the Simulation-Based Methodology

The choice of a simulation-based methodology is a deliberate engineering decision rather than a convenience. Applying controlled faults to a live 33/11 kV mining feeder would impose unacceptable safety, equipment, and operational risk, and the available infrastructure includes neither a real-time digital simulator (OPAL-RT or RTDS) nor a relay test bench or archive of mine-feeder fault records. Within these constraints, simulation is the only feasible method for controlled investigation. Specifically, the MATLAB/Simulink approach provides:

- **Safe investigation of severe faults.** High-current three-phase and near-bolted ground faults are applied in software, exposing neither personnel nor plant to fault energy.
- **Repeatable, controlled experiments.** Every scenario is deterministic and can be re-run identically, which is impossible with naturally occurring field faults.
- **Systematic parameter variation.** Fault type, location, resistance, loading level, and fault-inception time are swept independently across a defined grid.
- **A balanced, fully labelled dataset.** Each simulated case carries an exact ground-truth label, allowing a class-balanced dataset without manual annotation.
- **Observation of internal quantities.** Bus voltages and branch currents that would require extensive instrumentation in the field are directly available in simulation.
- **Lower cost and full reproducibility.** No field relay testing or outage is required, and the entire workflow can be regenerated from the archived scripts and model.

Simulation is used not as a substitute for field validation but as a means of establishing whether the proposed features, classifier, isolation logic, and restoration logic are technically coherent under controlled conditions. Its limitations are acknowledged from the outset: the data are synthetic; the sensors and communication links are idealised; the loading model is simplified; real relay and breaker operating delays are represented only as estimates rather than modelled hardware; there is no hardware-in-the-loop testing; and there is no field validation. A simulation-to-reality gap therefore remains, and the study is presented as a proof-of-concept design exercise that presupposes subsequent hardware-in-the-loop and field validation before any deployment. Simulation is not treated as equivalent to industrial deployment at any point in this thesis.

3.1.2 Mathematical Formulation

The mathematical formulation links the electrical network, the machine-learning classifier, and the restoration check within a single set of relationships, so that every claim can be verified against defined engineering variables. Each symbol is defined immediately after the equation in which it first appears.

Per-unit voltage. All bus voltages are interpreted in per-unit form so that different operating states can be compared against the same 0.95-1.05 pu band:

$$\boldsymbol{V_{pu}} = \frac{\boldsymbol{V_{RMS}}}{\boldsymbol{V_{base}}} \tag{3.1}$$

where V_{RMS} is the measured line-to-line RMS voltage at a monitored bus and V_{base} = 11 kV is the line-to-line base voltage on the feeder side.

Base current and base impedance. The base quantities used to sanity-check simulated currents and impedances follow from the base apparent power S_{base} and the base voltage:

$$I_{base} = \frac{S_{base}}{\sqrt{3} \cdot V_{base}}, \quad Z_{base} = \frac{(V_{base})^2}{S_{base}} \tag{3.2}$$

Three-phase apparent power. The apparent power measured at, or reconnected to, a bus is obtained from the line-to-line RMS voltage and line current:

$$S = \sqrt{3} \cdot V_{LL} \cdot I_L \tag{3.3}$$

where V_{LL} is the line-to-line RMS voltage and I_L the line RMS current. This relation converts the measured pre-fault voltage and current into the apparent power used by the restoration capacity check.

Source strength. The 33 kV grid is represented by a Thevenin equivalent whose impedance follows from the specified three-phase short-circuit level S_{sc} and the X/R ratio:

$$|Z_s| = \frac{(V_{base})^2}{S_{sc}}, \quad X_s = \frac{|Z_s|}{\sqrt{1 + \frac{1}{\left(\frac{X}{R}\right)^2}}}, \quad R_s = \frac{X_s}{\frac{X}{R}} \tag{3.4}$$

with S_{sc} = 500 MVA and X/R = 10. These values determine the fault-current level available at each bus and hence the strength of the current signatures on which classification depends.

Transformer loading ratio and capacity constraint. When the tie-switch reconnects load to transformer T2, the resulting loading ratio and the capacity constraint are:

$$\lambda_{T2} = \frac{S_{restored}}{S_{T2,rated}}, \quad \text{subject to } S_{restored} \leq S_{T2,rated} \tag{3.5}$$

where $S_{restored}$ is the total apparent power that transformer T2 must carry after restoration, which includes the load it already supplies at Bus B5, and $S_{T2,rated}$ = 15 MVA is the rated apparent power of T2. Because T2 continuously supplies B5, that existing load is counted: the capacity check sums the measured pre-fault apparent power of every bus that T2 will carry (the transferred buses together with B5), and restoration is permitted only when this total stays within the T2 rating.

Restored-voltage constraint. After any restoration action, the restored buses must remain inside the acceptable band:

$$0.95 \leq V_{pu} \leq 1.05 \tag{3.6}$$

Feature vector. The classifier does not use raw waveforms. Each phase voltage and current is reduced to an RMS value over a finite window, and the per-bus values are assembled into a 24-element vector for each case i:

$$x = [V_{B2,AB}\ V_{B2,BC}\ V_{B2,CA}\ I_{B2,A}\ I_{B2,B}\ I_{B2,C}\ \cdots\ V_{B5,AB}\ V_{B5,BC}\ V_{B5,CA}\ I_{B5,A}\ I_{B5,B}\ I_{B5,C}]^T$$

(3.7)

where the logged RMS voltages are line-to-line quantities (V_{AB}, V_{BC}, V_{CA}) and the currents are line currents (I_A, I_B, I_C) at buses B2, B3, B4 and B5. The vector encodes a spatial electrical signature across the feeder, not a single relay measurement.

Dataset size. The dataset size follows directly from the experimental design, four fault locations, three fault types, five resistance values, five loading levels and three onset times for the fault classes, with the healthy class generated separately:

$$\boldsymbol{N} = \boldsymbol{N_{healthy}} + \left(\textbf{locations} \times \textbf{types} \times \boldsymbol{R_f} \times \textbf{LM} \times \boldsymbol{t_{on}}\right) = \mathbf{100} + (\mathbf{4} \times \mathbf{3} \times \mathbf{5} \times \mathbf{5} \times \mathbf{3}) = \mathbf{1000} \tag{3.8}$$

The 1,000 samples are thus a documented consequence of the parameter sweep rather than an arbitrary choice. The 75 samples per fault class are the full factorial grid of the swept parameters (five fault resistances × five load multipliers × three onset times = 75), so no subsampling is applied.

Every one of the 900 fault samples is therefore a distinct parameter combination. The 80/20 partition of Section 3.4.1 is applied per sample with class stratification (fixed seed rng = 42), so no sample appears in both the training and test sets. The 100 healthy samples are generated at ten load levels spanning 0.55–1.35 pu; because the model is deterministic, repeated runs at a given level coincide, so the healthy class effectively spans ten distinct operating points — a data-diversity limitation that added measurement noise would remove, as examined in Section 5.7.

Random Forest classification. The predicted class is produced by the trained forest, and equivalently by majority vote across its M trees:

$$\hat{y} = \boldsymbol{f_{RF}}(\boldsymbol{x}) = \textbf{mode}\{\boldsymbol{h_m}(\boldsymbol{x})\},\ \ \boldsymbol{m} = \mathbf{1}, \dots, \boldsymbol{M},\ \ \boldsymbol{M} = \mathbf{500} \tag{3.9}$$

where f_{RF} is the trained classifier, h_m the m-th decision tree, $\hat{y} \in \{0,\dots,12\}$ the predicted class (0 = Healthy), and M the number of trees. The number of predictors sampled at each split uses the square-root rule for classification forests, $m_{try} = \lfloor\sqrt{d}\rfloor = \lfloor\sqrt{24}\rfloor = 4$.

Cost-sensitive risk. Training minimises a cost-weighted empirical risk in which predicting Healthy for a true fault is penalised more heavily than any other error:

$$\boldsymbol{R_{cost}} = \sum_{\boldsymbol{i}} \boldsymbol{C}(\boldsymbol{y_i}, \hat{\boldsymbol{y}}_{\boldsymbol{i}}),\ \ \boldsymbol{C}(\textbf{fault} \rightarrow \textbf{Healthy}) = \mathbf{12.5},\ \ \textbf{otherwise}\boldsymbol{C} = \mathbf{1} \tag{3.10}$$

Classification metrics. Performance is measured by accuracy and by per-class precision, recall and F1-score, so that weak performance on one fault class cannot be hidden behind a high overall figure:

$$\textbf{Accuracy} = \frac{\textbf{correct predictions}}{\textbf{total predictions}} \tag{3.11}$$

$$\textbf{Precision}_{\boldsymbol{i}} = \frac{\textbf{TP}_{\boldsymbol{i}}}{\textbf{TP}_{\boldsymbol{i}} + \textbf{FP}_{\boldsymbol{i}}},\ \ \textbf{Recall}_{\boldsymbol{i}} = \frac{\textbf{TP}_{\boldsymbol{i}}}{\textbf{TP}_{\boldsymbol{i}} + \textbf{FN}_{\boldsymbol{i}}},\ \ \textbf{F1}_{\boldsymbol{i}} = \frac{\mathbf{2} \cdot \textbf{Precision}_{\boldsymbol{i}} \cdot \textbf{Recall}_{\boldsymbol{i}}}{\textbf{Precision}_{\boldsymbol{i}} + \textbf{Recall}_{\boldsymbol{i}}}$$

(3.12)

where TP_i, FP_i and FN_i are the true positives, false positives and false negatives for class i.

Restoration decision. The tie-switch is not closed simply because a fault has been found. The tie-switch command is:

$$\boldsymbol{u_{tie} = 1} \ \ \textbf{if all restoration conditions hold}, \ \ \textbf{otherwise } \boldsymbol{u_{tie} = 0} \tag{3.13}$$

This makes the self-healing action conditional: it is restoration only when supply availability, transformer capacity, and voltage acceptability are simultaneously satisfied, and it is withheld for terminal-zone faults where closing the tie would backfeed the fault.

Service-continuity metrics. The operational benefit is quantified by the number of healthy zones left de-energised after switching and, for restored buses, by the voltage recovery ratio:

$$\boldsymbol{N_{healthy,off}} = \textbf{number of healthy zones de-energised after switching} \tag{3.14}$$

$$\boldsymbol{R_V = \frac{V_{post}}{V_{pre-fault}}} \tag{3.15}$$

where V_{post} is the restored steady-state voltage and $V_{pre\text{-}fault}$ the healthy voltage at the same bus. $N_{healthy,off}$ is the primary service-continuity measure used in the comparative benchmark of Chapter 5.

3.2 Feeder Model Specification

3.2.1 Network Topology

The feeder model represents a 33/11 kV mining distribution network with five buses and two supply transformers. Figure 3.2 gives the block-level topology and Figure 3.3 the single-line diagram.

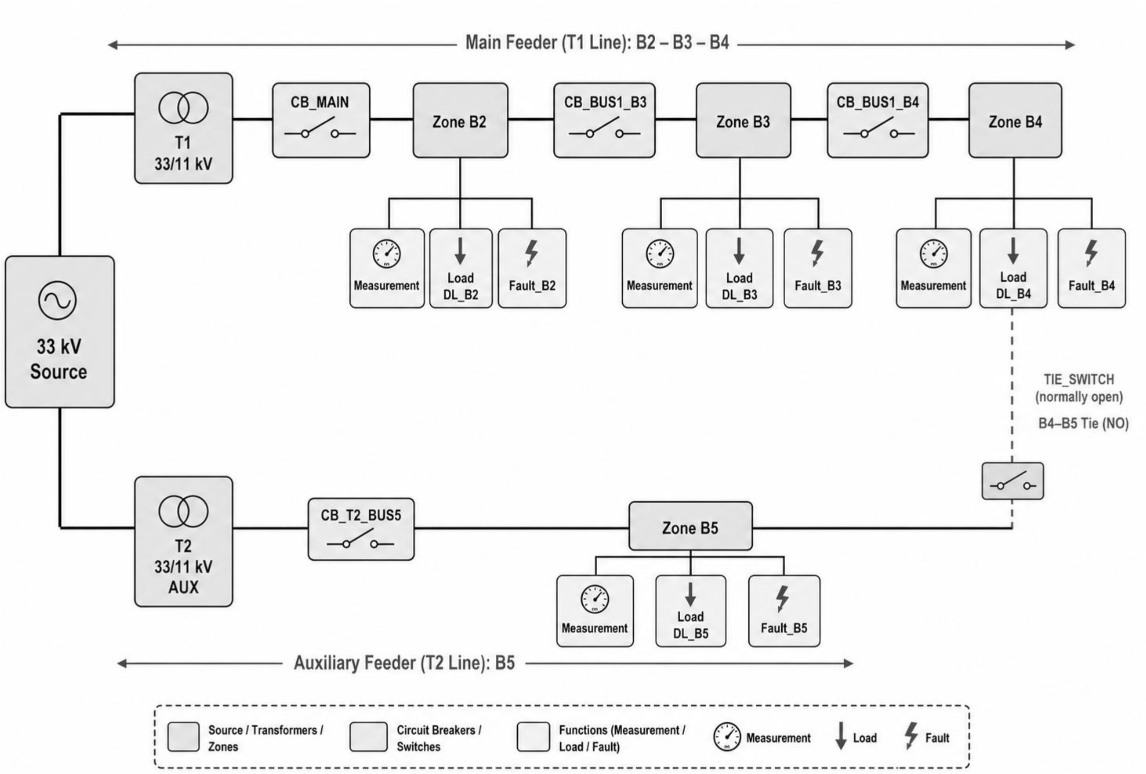

Figure 3.2: Feeder topology block diagram

The block diagram reduces the feeder to the elements relevant to this study, namely T1, T2, the load buses, section breakers, tie-switch, and fault locations, rather than reproducing a formal protection single-line drawing. This abstraction also exposes the central constraint on restoration: because the feeder is radial with a single tie-switch, only sections with a safe alternative path through the B5/T2 side can be re-energised after isolation.

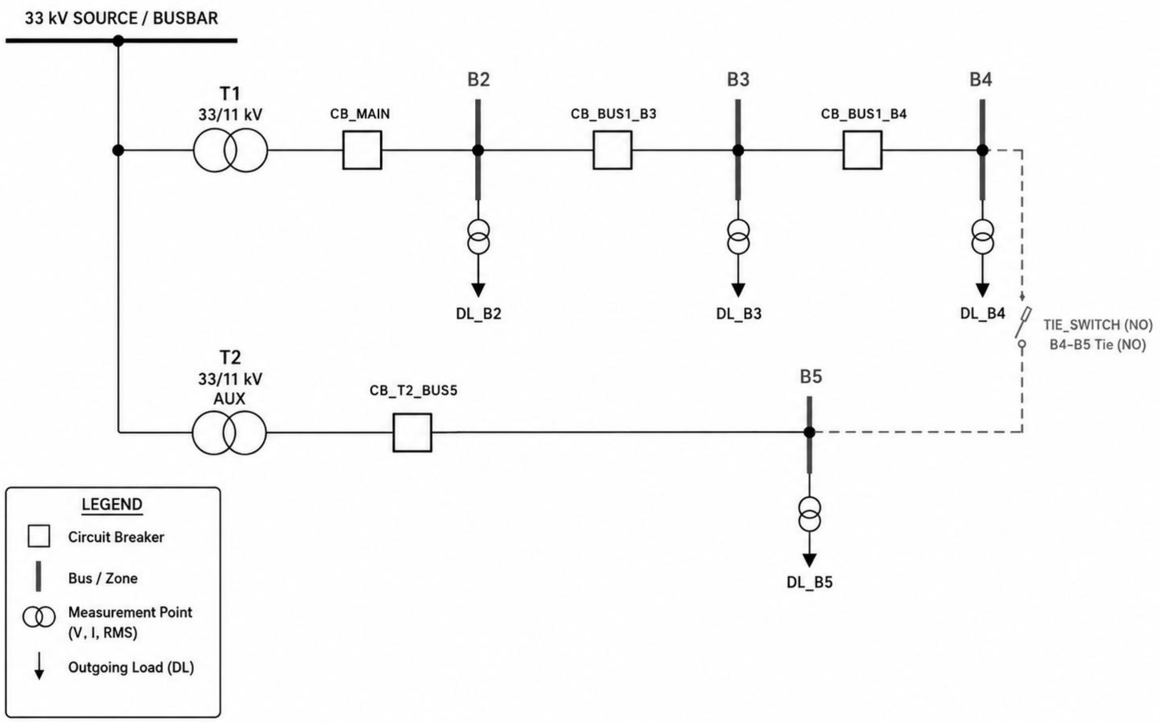

Figure 3.3: Single-line diagram of the five-bus feeder

The feeder has two distinct supply paths. The main path originates at transformer T1 (33/11 kV, 20 MVA), which supplies the series-radial 11 kV feeder through the main breaker CB_MAIN and the section breakers CB_BUS1_B3 and CB_BUS1_B4 to buses B2 (dewatering), B3 (ventilation) and B4 (crusher). The alternative path runs from transformer T2 (33/11 kV, 15 MVA) through CB_T2_BUS5 to Bus B5 (SX-EW plant), which is continuously energised from T2 in all operating states. The normally-open tie-switch TIE_SWITCH links Bus B4 and Bus B5 to enable post-fault restoration. Figure 3.4 shows the resulting protection zones.

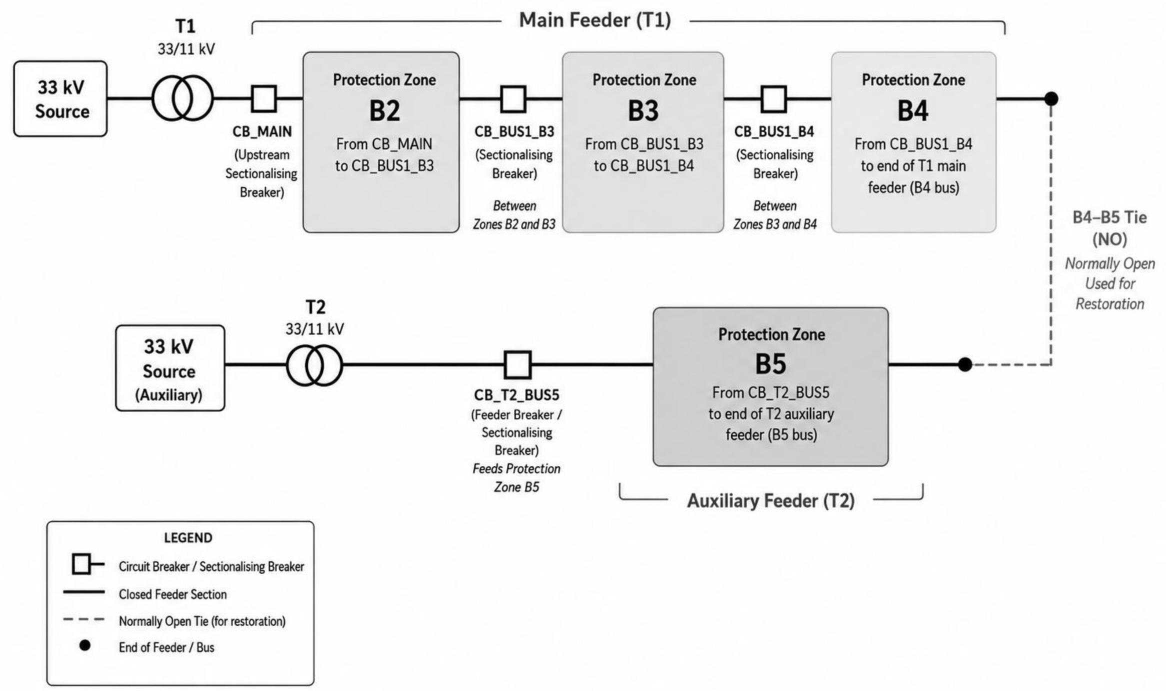

Figure 3.4: Protection zones and the normally-open tie

Each load bus defines one protection zone monitored by a three-phase V-I measurement carrying that bus's protected branch. Feeder continuation taps upstream of each measurement, so a fault in a zone drives fault current through that zone's own measurement, giving a localised current signature.

3.2.2 Load Specification

The four process loads define the feeder operating point. Their ratings, listed in Table 3.1, set the loading against which voltage drop, transformer utilisation, and restoration feasibility are assessed throughout the study. The loads are modelled as three-phase series RLC loads at 11 kV nominal, 50 Hz, specified by active power at unity power factor.

Table 3.1: Load-centre specifications

Bus	Process load	Active power P (MW)	Reactive power Q (MVAr)
B2	Underground dewatering pumps	1.950	0 (unity pf)
B3	Mine ventilation systems	2.600	0 (unity pf)
B4	Crusher and conveyor systems	3.250	0 (unity pf)
B5	SX-EW plant (T2 supply)	2.145	0 (unity pf)
Total		9.945	0

Note: Values are the nominal (LM = 1.00) active powers configured in the DL_B2-DL_B5 blocks of the final model; the parameter sweep scales these by the load multiplier LM ∈ [0.70, 1.30]. The nominal level was verified against the unscaled healthy pre-flight currents (e.g. B2 = 103.7 A, matching the 1.95 MW full-load current of 102.3 A). The load blocks specify active power only; reactive power is zero (unity power factor) as modelled. A field installation would typically operate near 0.95 lagging, which would raise load currents modestly and is noted as a modelling simplification in Section 6.7.

3.2.3 Source and Transformer Parameters

The 33 kV supply is modelled as a Thevenin equivalent with a three-phase short-circuit level of 500 MVA and X/R ratio of 10 at 50 Hz, representing a strong transmission grid [19]. Both transformers use a Dyn11 vector group, that is a 33 kV delta primary (D11) and a solidly earthed 11 kV star secondary (Yg), with per-unit leakage of 0.002 pu resistance and 0.04 pu inductance per winding (approximately 8% impedance). Transformer T1 is rated 20 MVA and transformer T2 is rated 15 MVA. The full numerical parameter set, as read from the model, is documented in Chapter 4 (Tables 4.1-4.3).

3.3 Fault Simulation and Dataset Generation

3.3.1 Fault Types and Phase Configuration

Three fault types are simulated: single line-to-ground (SLG), where Phase A contacts earth through the fault resistance with the ground-fault path enabled; line-to-line (LL), where Phases A and B are short-circuited; and three-phase balanced (3PH), where all three phases are simultaneously short-circuited. These types span the range from the most asymmetric (SLG) to the most symmetric (3PH) condition and represent the dominant fault modes in medium-voltage distribution systems.

3.3.2 Parameter Sweep Design

Table 3.2 lists the swept parameters, fault resistance, load level, and fault onset time, that vary each scenario about the nominal condition. The sweep broadens the dataset beyond a single ideal case while keeping the parameter space bounded and reproducible.

Table 3.2: Fault parameter sweep values

Parameter	Levels	Physical interpretation
Fault resistance R_f (Ω)	0.001 / 0.1 / 0.5 / 1.0 / 5.0	Near-metallic to moderate high-impedance
Load multiplier LM (pu)	0.70 / 0.85 / 1.00 / 1.10 / 1.30	Light to heavy loading conditions
Fault onset time t_{on} (s)	0.50 / 0.75 / 1.00	Distribution within the simulation window

3.3.3 Class Structure

Table 3.3 maps each numerical class label to its electrical meaning, healthy operation or a specific fault type and location, and records the stratified 80/20 split, which guarantees that every class is represented in both the training and test partitions.

Table 3.3: Class labels and stratified train-test split

Class	Label	Fault type	Location	n train	n test
0	Healthy	No fault, system normal	System-wide	80	20
1	SLG-B2	Single line-to-ground	Bus B2	60	15
2	LL-B2	Line-to-line (A-B)	Bus B2	60	15
3	3PH-B2	Three-phase balanced	Bus B2	60	15
4	SLG-B3	Single line-to-ground	Bus B3	60	15
5	LL-B3	Line-to-line (A-B)	Bus B3	60	15
6	3PH-B3	Three-phase balanced	Bus B3	60	15
7	SLG-B4	Single line-to-ground	Bus B4	60	15
8	LL-B4	Line-to-line (A-B)	Bus B4	60	15
9	3PH-B4	Three-phase balanced	Bus B4	60	15
10	SLG-B5	Single line-to-ground	Bus B5	60	15
11	LL-B5	Line-to-line (A-B)	Bus B5	60	15
12	3PH-B5	Three-phase balanced	Bus B5	60	15
Total	13 classes			800	200

3.3.4 Feature Extraction

For each simulation run, three-phase RMS voltage and current signals are logged to the MATLAB workspace. The disturbance onset is detected from the measured signals themselves, the point at which current exceeds 1.5 times its pre-fault baseline or voltage falls below 0.85 pu, and each feature is taken as the mean of the corresponding RMS signal over a short window in the post-onset steady state. This data-driven timing ensures that the feature sample reflects the settled fault condition without assuming prior knowledge of the injection time. No normalisation is applied, because the Random Forest is scale-invariant.

3.4 Classifier Design

3.4.1 Train-Test Partition

The 1,000-sample dataset is partitioned 80/20 by stratified random sampling (rng = 42) into 800 training and 200 test samples, the stratification preserving the class proportions at 20 Healthy and 15 per fault class in the test set. The test set is withheld before training and reserved for the final evaluation in Chapter 5. To reduce dependence on any single partition, five-fold stratified cross-validation is reported as a supplementary check.

3.4.2 Random Forest Configuration and Cost Matrix

The classifier is a 500-tree Random Forest [17]. Each tree is grown on a bootstrap sample (approximately 63.2% of the training data) to full depth (minimum leaf size = 1), sampling $\lfloor\sqrt{24}\rfloor = 4$ features per split, with the random seed fixed at rng = 42 for reproducibility. A 13×13 asymmetric misclassification cost matrix assigns weight 12.5 to any prediction of Healthy when the true class is a fault (a false negative); all other off-diagonal errors carry unit weight and correct predictions carry zero cost. This asymmetry reflects the operational priority of mining protection: a missed fault may sustain unsafe conditions, whereas a false alarm triggers a reversible isolation action. The training pipeline is shown in Figure 3.5.

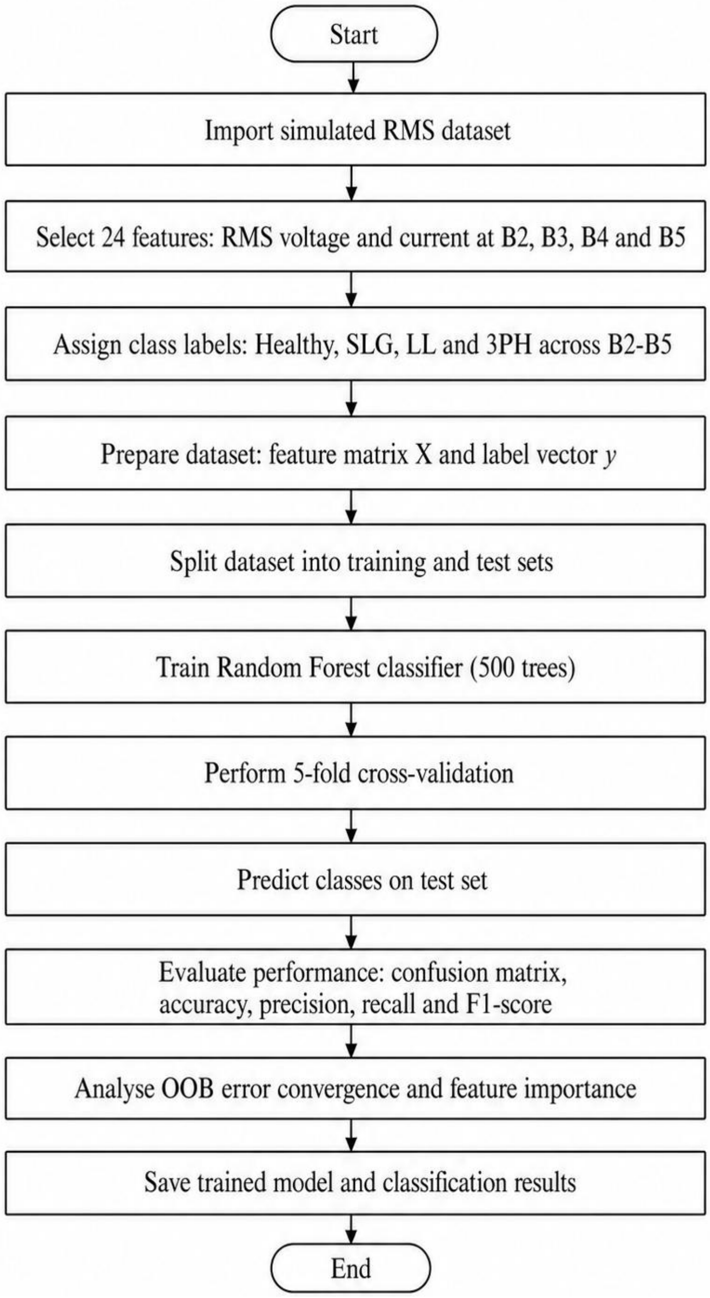

Figure 3.5: Machine-learning training pipeline

Documenting this sequence, from data loading and stratified splitting through cost-sensitive fitting to held-out evaluation, makes the reported accuracy reproducible and directly attributable to a defined training procedure.

3.5 Selective Isolation Logic

The isolation logic maps the Random Forest prediction to a breaker command through a location-based rule. A prediction of Healthy produces no action. A predicted fault at B2 opens CB_MAIN and the downstream section breaker CB_BUS1_B3 to isolate the B2 section; a predicted fault at B3 opens CB_BUS1_B3 and CB_BUS1_B4; a predicted fault at B4 opens CB_BUS1_B4; and a predicted fault at B5 opens CB_T2_BUS5. Fault-type misclassifications within the same bus group still produce the correct isolation action, because the breaker is selected by predicted location, not fault type. Crucially, the command is issued from the classifier prediction computed on the pre-switch measurements, so the evaluation forms a genuine closed loop rather than a look-up against the known fault. Figure 3.6 defines the mapping.

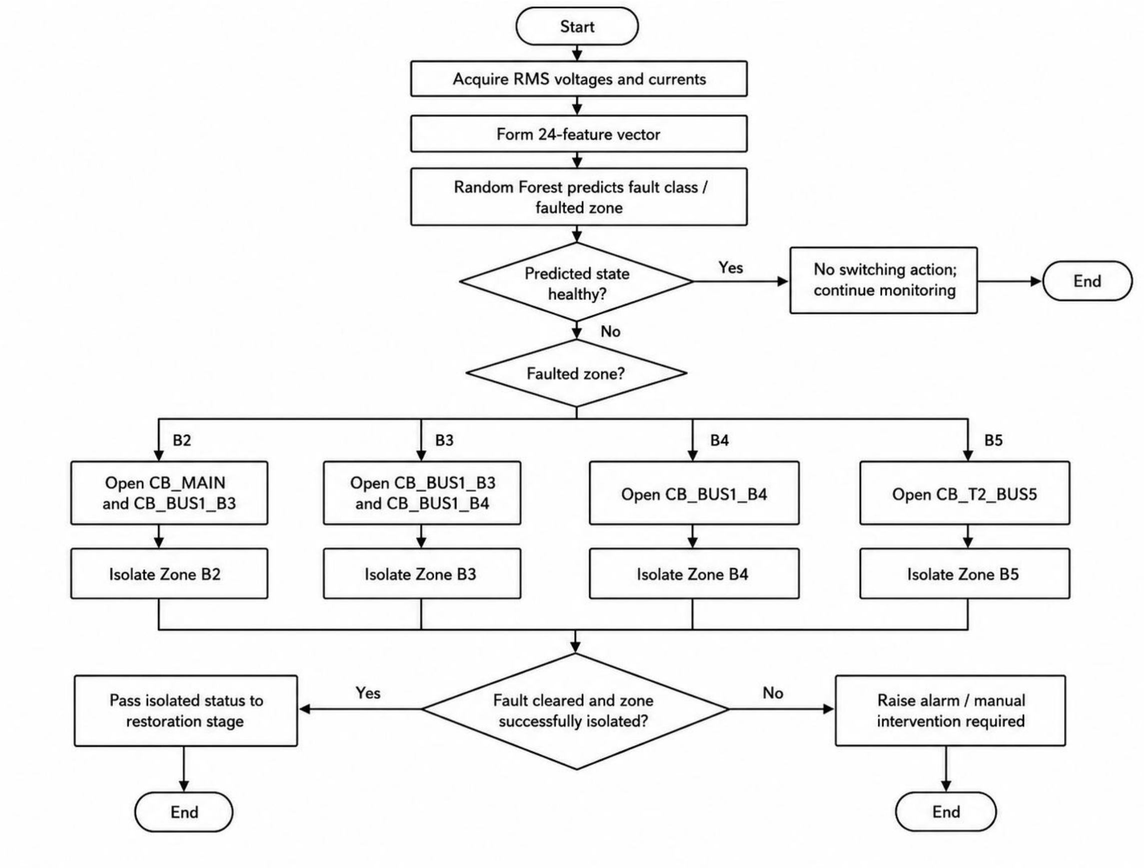

Figure 3.6: Selective fault-isolation logic

For a series-radial feeder, opening CB_BUS1_B3 to clear a B3 fault also de-energises Bus B4 downstream; the tie-switch restores B4 via T2 while Bus B3 remains de-energised until manual crew intervention. Similarly, a B2 fault de-energises B2, B3 and B4, and the tie restores the downstream healthy buses that lie beyond the isolation point. Complete restoration of all healthy upstream buses is not achievable with a single tie-switch on a series-radial topology; this topological limitation is acknowledged in Section 1.5 and revisited in Chapter 6.

3.5.1 Conventional Relay Coordination Benchmark

A conventional overcurrent benchmark is retained so that the proposed ML-assisted isolation can be compared with a familiar protection-engineering reference. The benchmark uses the IEC standard inverse IDMT characteristic [18] with a coordination time interval of approximately 0.20 s to represent relay processing, breaker clearing and grading allowance. The settings in Table 3.4 are illustrative values based on the simulated current range and are used for comparison, not as final relay settings for a real mine feeder.

Table 3.4: Illustrative conventional relay coordination benchmark

Relay position	Pickup (A)	TMS	Time at 4.5 kA (s)	Coordination role
B4 section relay	200	0.10	0.218	Primary for terminal B4 faults
B3 upstream relay	250	0.18	0.423	Backup to B4; CTI ≈ 0.205 s
B2 upstream relay	300	0.28	0.704	Second backup; CTI ≈ 0.281 s
Source-side relay	400	0.40	1.129	Final upstream backup; CTI ≈ 0.425 s

Note: Illustrative IDMT settings for comparison only; the benchmark determines the order of tripping and provides no information for post-fault restoration.

3.6 Self-Healing Restoration Logic

Following confirmed fault isolation, the restoration logic evaluates three conditions before closing the normally-open tie-switch. First, T2 must be healthy and energised. Second, the total reconnected load must not exceed the 15 MVA rating of T2. The reconnected apparent power is estimated from the measured pre-fault active power of the buses to be restored using Equation (3.3), and each bus's measurement is independent, so the per-bus apparent powers are summed. Third, the restored buses must satisfy the 0.95-1.05 pu voltage band. If any condition fails, or if the fault lies in a terminal zone (B4 or B5) where closing the tie would backfeed the fault, the tie-switch remains open, and the healthy buses stay on their existing supply. These checks act at different points in the sequence. The capacity check is a pre-switch estimate: it uses the apparent power measured on the intact network before isolation (Stage 1), summed over the buses that T2 will carry — the transferred buses together with the pre-existing B5 load — so the decision is available before the tie is closed. The voltage check is, by contrast, a post-switch validation: the restored-bus voltages are read after the tie has been closed (Stage 2), and the tie is reopened if any restored bus falls outside the 0.95–1.05 pu band. The autonomous in-model controller of Section 4.8 applies only the pre-switch capacity check, which the 15 MVA rating of T2 amply satisfies for this feeder; a deployable implementation would add a predictive load-flow estimate of the post-restoration voltage before closure. The decision is formalised in Equation (3.13) and illustrated in Figure 3.7.

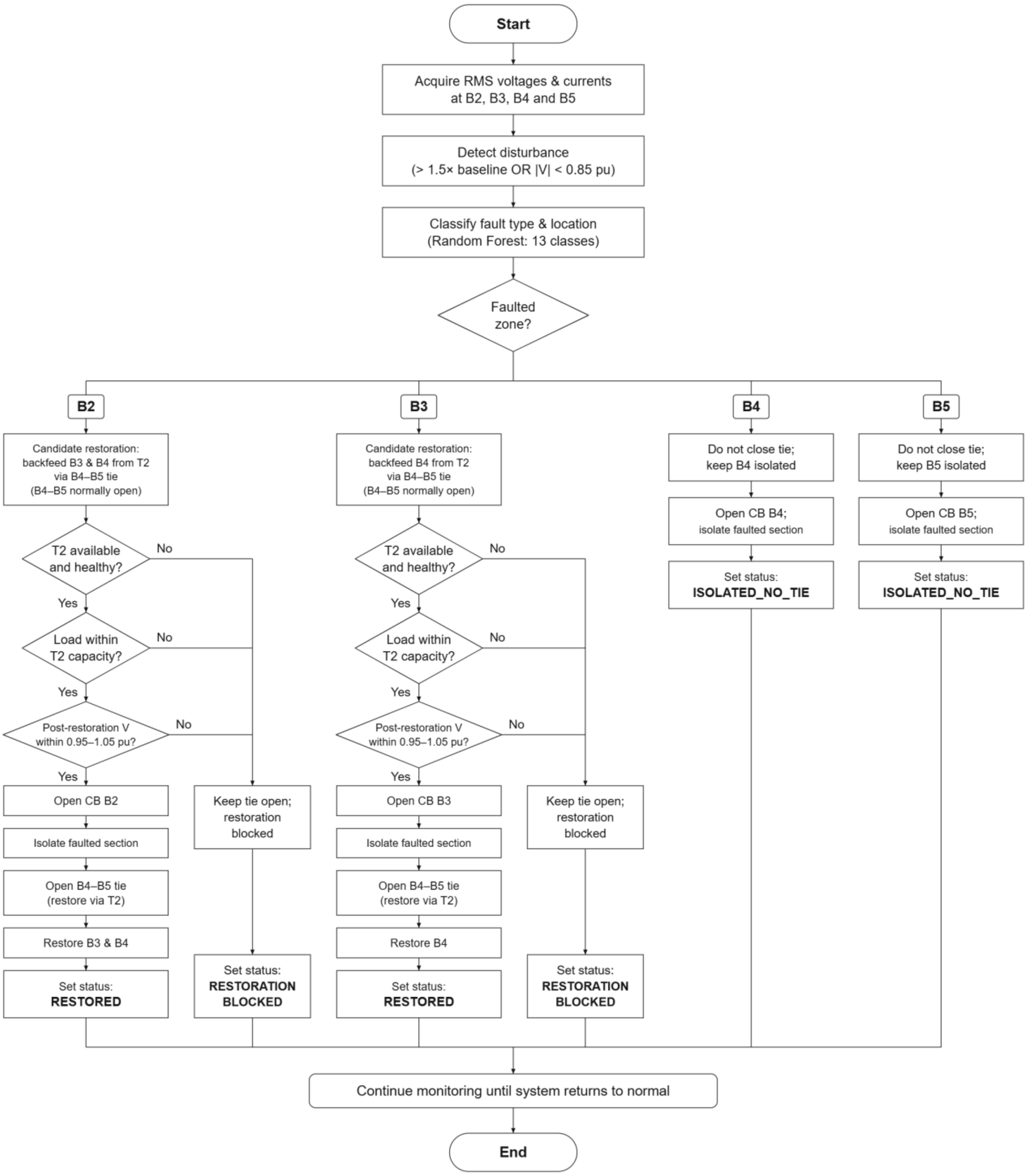

Figure 3.7: Self-healing restoration logic

The restoration logic is presented as a conditional control action rather than unconditional switching: the tie-switch closes only after supply availability, transformer capacity, and voltage limits have all been verified. These interlocks keep the self-healing claim bounded by physical constraints and guarantee that the tie is never closed into an un-cleared fault.

3.7 Ethical, Safety, and Data-Integrity Considerations

3.7.1 Ethical Considerations

- No human participants were involved, and no personal or confidential mine data were used; all data were generated from the Simulink model.
- No field protection equipment was accessed or manipulated at any stage.
- The synthetic nature of the data is reported transparently, and no claim of industrial readiness or field validation is made.

3.7.2 Safety Considerations

- Simulation avoids exposing personnel and equipment to fault currents; all switching commands are evaluated virtually.
- Any actual deployment would require review by a qualified protection engineer, and real operation would require interlocks, synch-check where applicable, grounding checks, relay coordination, and utility/mine operating approval.
- The proposed tie closure is constrained so that it can never energise an un-cleared fault; terminal-zone faults keep the tie open by design.
- Field implementation would additionally require fail-safe behaviour and manual-override provisions.

3.7.3 Data-Integrity Considerations

- Scenario generation is deterministic and documented, with a fixed random seed (rng = 42) for repeatable splitting and training.
- Class labels are assigned consistently, and the training and test sets are separated before any training activity.
- Raw simulation outputs are preserved, and traceability is maintained between scripts, dataset, result summaries, figures, and tables.
- Failed and blocked scenarios (for example terminal-zone faults where restoration is withheld) are retained rather than discarded.
- Units and per-unit conversions are verified, and the final scripts, model, dataset, and outputs are stored so that the workflow is reproducible.
- No results were manually altered; every reported figure is regenerated from the stored outputs.

3.8 Chapter Summary

This chapter converted the research idea into a measurable methodology. It justified the simulation-based approach and stated its limitations, formalised the electrical, machine-learning and restoration relationships in an explicit set of equations, and defined the feeder topology, the fault sweep, the 24-feature RMS input vector, the cost-sensitive Random Forest, the selective isolation rule, the conditional restoration logic, and the ethical, safety and data-integrity safeguards. These elements connect the electrical behaviour, the classifier decision, and the restoration action so that the proposed scheme is supported by engineering theory, not only by simulation output.

Chapter 4 System Design and Implementation

4.1 Introduction

This chapter documents the translation of the methodology of Chapter 3 into the working MATLAB R2024a Simulink environment. It presents the implemented feeder model, the block parameters read directly from the model (transcribed here as editable tables), the healthy-state validation, the grounding and measurement configuration that ensures physically correct single line-to-ground fault currents, and the Random Forest training configuration.

4.2 Simulink Feeder Model

The mining feeder is implemented as mining_feeder_layer_FINAL_baseline.slx using the Simscape Electrical Specialized Power Systems library. The model runs under the powergui block in discrete mode with a fixed sample time of 5 µs (200 kHz); all RMS measurement and logging blocks execute at this rate. Each run simulates 2.0 s of feeder operation. Figure 4.1 shows the complete model canvas.

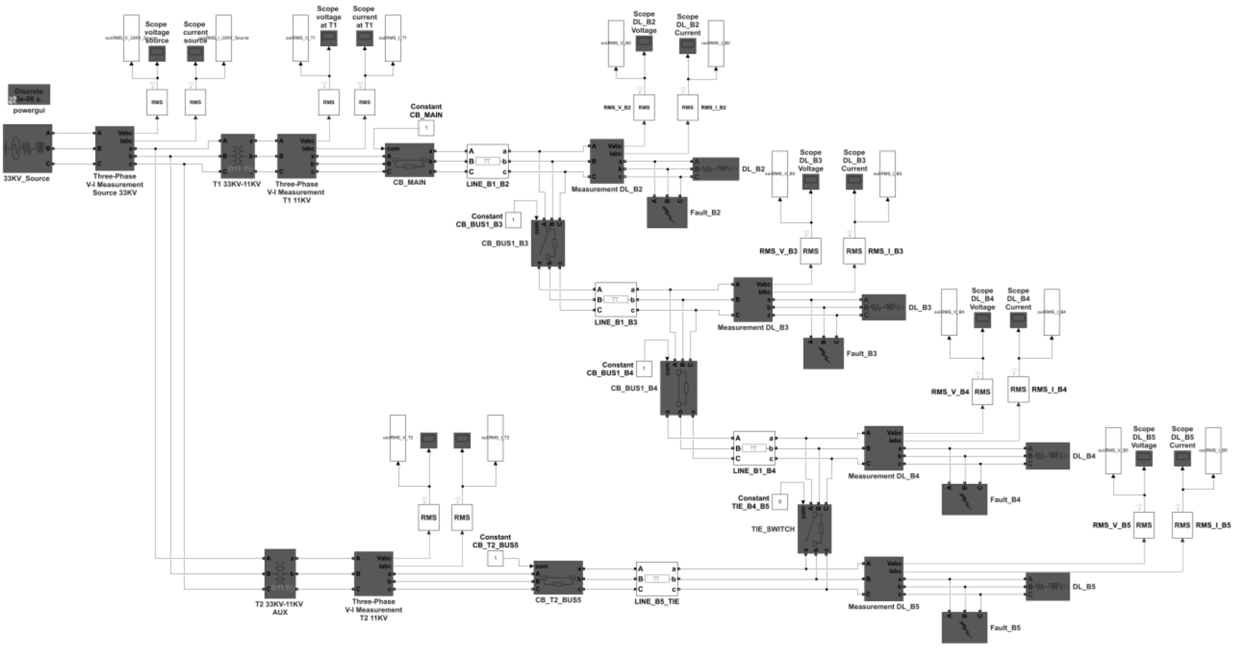

Figure 4.1: Simulink feeder model canvas

The model integrates the 33 kV source, transformers T1 and T2, the radial 11 kV sections, the auxiliary supply, the fault blocks, the per-bus RMS measurements, and the four load centres in a single layout, establishing the correspondence between the methodology and the executed model. The parameter values configured in each block group are transcribed from the model in Tables 4.1-4.8.

4.2.1 Source Parameters

Table 4.1: 33 kV source parameters

Parameter	Value
Nominal voltage (line-to-line RMS)	33 kV
Frequency	50 Hz

Internal connection	Solidly earthed star (Yg)
Three-phase short-circuit level	500 MVA
X/R ratio	10
Equivalent source impedance (33 kV)	Z_s = 2.178 Ω (R_s = 0.217 Ω, X_s = 2.167 Ω)
Short-circuit current at 33 kV bus	≈ 8.75 kA

Note: Derived quantities (Z_s, short-circuit current) computed from the nameplate short-circuit level and X/R using Equation (3.4).

4.2.2 Transformer Parameters

Table 4.2: Transformer T1 parameters

Parameter	Value
Rating	20 MVA
Voltage ratio	33 / 11 kV
Vector group	Dyn11 (delta D11 primary / earthed-star Yg secondary)
Leakage impedance per winding	0.002 + j0.040 pu (≈ 8% total)
Magnetising resistance / inductance	Rm = 499.98 pu, Lm = 500 pu
Full-load current (33 kV / 11 kV)	349.9 A / 1049.7 A

Table 4.3: Transformer T2 parameters

Parameter	Value
Rating	15 MVA
Voltage ratio	33 / 11 kV
Vector group	Dyn11 (delta D11 primary / earthed-star Yg secondary)
Leakage impedance per winding	0.002 + j0.040 pu (≈ 8% total)
Magnetising resistance / inductance	Rm = 499.98 pu, Lm = 500 pu
Full-load current (33 kV / 11 kV)	262.4 A / 787.3 A
Role	Auxiliary supply to Bus B5 and tie-switch restoration source

4.2.3 Line Parameters

Table 4.4: Distribution line (pi-section) parameters

Line	Route	Length (km)	R1 / R0 (Ω/km)	L1 / L0 (mH/km)	C1 / C0 (µF/km)
LINE_B1_B2	Main → B2	0.8	0.12 / 0.36	0.35 / 1.05	0.20 / 0.10
LINE_B1_B3	Main → B3	1.2	0.12 / 0.36	0.35 / 1.05	0.20 / 0.10
LINE_B1_B4	Main → B4	1.0	0.12 / 0.36	0.35 / 1.05	0.20 / 0.10
LINE_B5_TIE	T2 → B5 / tie	0.6	0.12 / 0.36	0.35 / 1.05	0.20 / 0.10

Note: Subscript 1 = positive-sequence, 0 = zero-sequence. All four lines share identical per-kilometre parameters; only the route length differs.

4.2.4 Load Block Parameters

Table 4.5: Load block parameters

Load block	Nominal voltage	Frequency	Configuration	Active power
DL_B2	11 kV	50 Hz	3-phase series RLC	1.950 MW
DL_B3	11 kV	50 Hz	3-phase series RLC	2.600 MW
DL_B4	11 kV	50 Hz	3-phase series RLC	3.250 MW
DL_B5	11 kV	50 Hz	3-phase series RLC	2.145 MW

4.2.5 Circuit-Breaker and Tie-Switch Parameters

Table 4.6: Circuit-breaker and tie-switch parameters

Device	Normal state	On-resistance (Ω)	Snubber (R ‖ C)	Control
CB_MAIN	Closed	0.01	1 MΩ ‖ inf	External
CB_BUS1_B3	Closed	0.01	100 kΩ ‖ 1 nF	External
CB_BUS1_B4	Closed	0.01	100 kΩ ‖ 1 nF	External
CB_T2_BUS5	Closed	0.01	100 kΩ ‖ 1 nF	External
TIE_SWITCH	Open (N/O)	0.01	100 kΩ ‖ 1 nF	External

Note: Breakers are driven by external Constant blocks (1 = closed, 0 = open). The scripts set the normal operating state, all section breakers closed and the tie open, at the start of every run.

4.2.6 Fault-Block Parameters

Table 4.7: Fault-block parameters

Parameter	Value
Fault blocks	Fault_B2, Fault_B3, Fault_B4, Fault_B5
Fault (on) resistance R_{on}	0.001 Ω (swept 0.001-5.0 Ω)
Ground resistance R_g	0.001 Ω (valid ground-return path)
Snubber	100 kΩ ‖ 1 nF
Fault types applied	SLG (A-g), LL (A-B), 3PH (A-B-C), set programmatically

Note: Fault type and ground-fault flags are applied at run time by the MASTER_A script; the ground resistance is set to 0.001 Ω on all four blocks so that SLG faults present a physically valid low-impedance earth return.

4.2.7 Measurement and Solver Configuration

Table 4.8: RMS measurement and solver configuration

Parameter	Value
powergui simulation mode	Discrete
Fixed sample time	5 µs (200 kHz)
RMS measurement	Per-phase true RMS at fundamental 50 Hz
Measurement placement	In series with each zone's protected branch (relay-side

	CT)
Logged quantities	3-phase RMS V and I at B2, B3, B4, B5 (24 signals)
Simulation duration per run	2.0 s
Environment	MATLAB R2024a + Simscape Electrical Specialized Power Systems

4.3 Healthy-State Validation

Before generating the fault matrix, the feeder was validated under healthy conditions. The RMS voltages and currents settle to balanced, steady values consistent with the specified loads and transformer ratings. Figures 4.2-4.4 show representative healthy outputs at the source and at the two restoration-relevant buses B4 and B5; the remaining buses were verified in the same way.

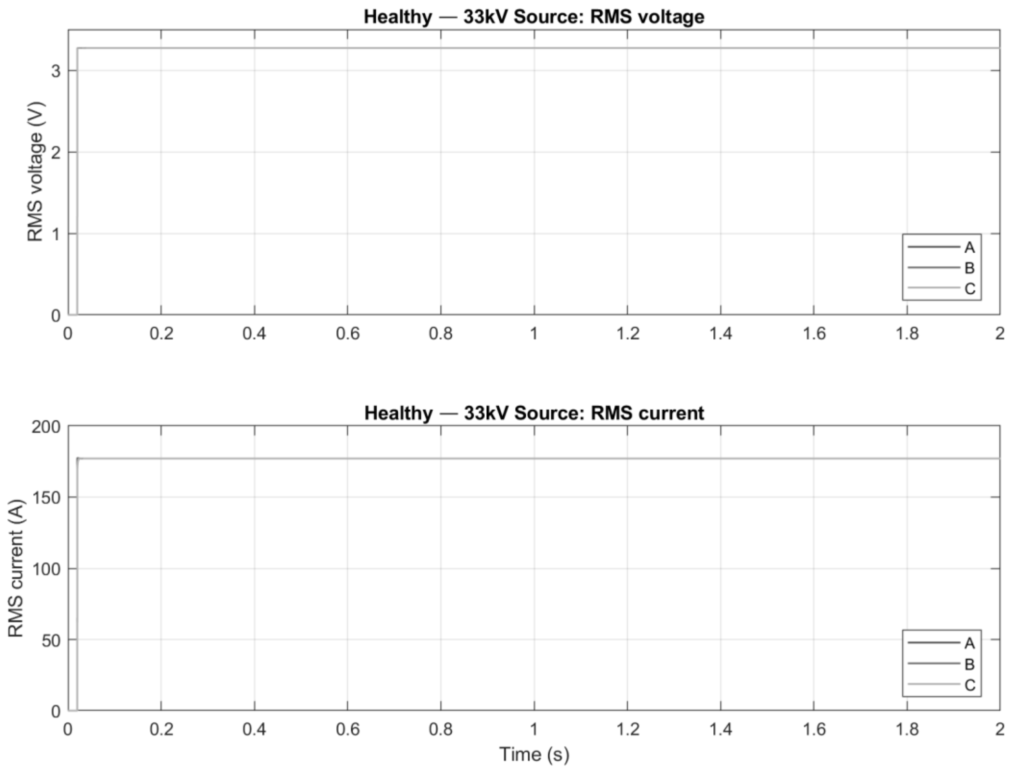

Figure 4.2: Healthy RMS voltage and current at the 33 kV source

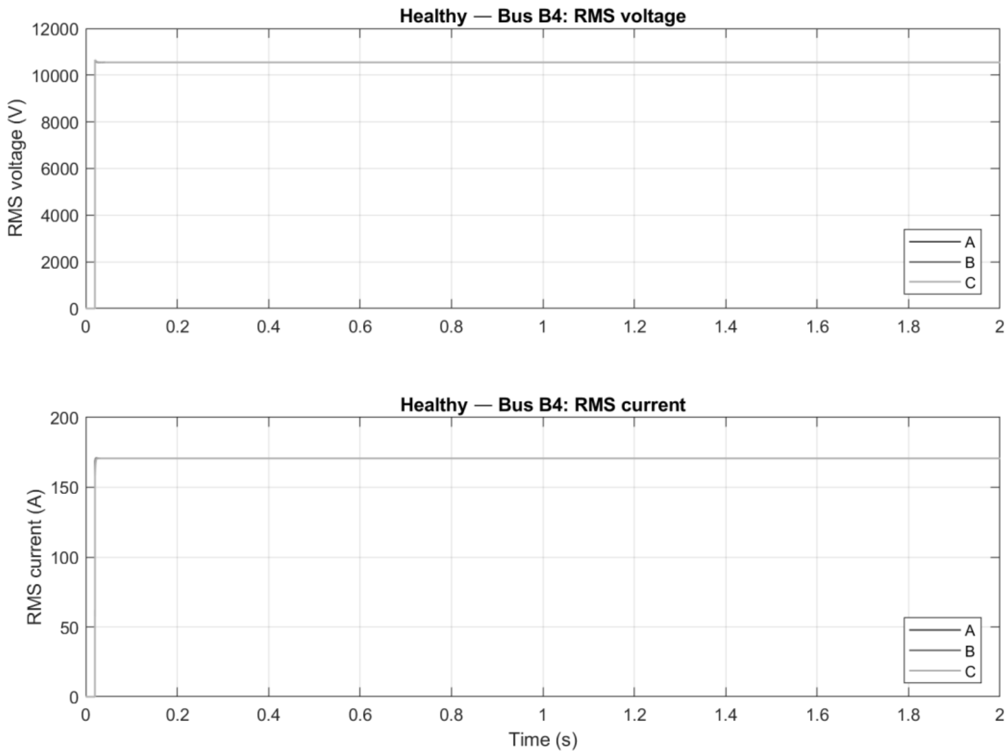

Figure 4.3: Healthy RMS voltage and current at Bus B4

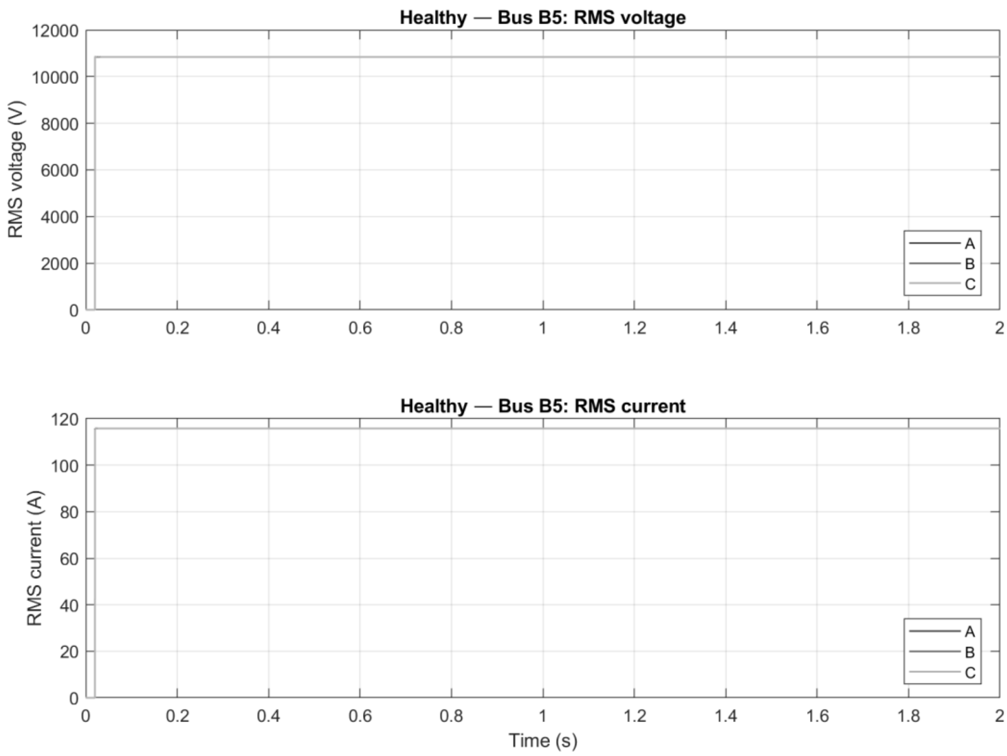

Figure 4.4: Healthy RMS voltage and current at Bus B5 (T2 supply)

A correct upstream and per-bus baseline is a precondition for interpreting the subsequent fault signatures, which would otherwise be confounded by an incorrectly energised model. The measured healthy currents also provide a quantitative check against analytical load flow, presented in Section 4.5.

4.4 Grounding and Measurement Verification (SLG Pre-flight)

Single line-to-ground faults are only informative for classification if the ground-fault current actually flows through the zone measurement. The model was therefore configured so that each three-phase measurement sits in series with its own fault-and-load branch, and the ground resistance of every fault block is set to a valid low value (0.001 Ω). To guarantee this configuration before any data are generated, the MASTER_A script runs a strict SLG grounding pre-flight: it applies an SLG fault in each zone and confirms that the faulted-phase current rises sharply above the healthy baseline, aborting dataset generation if any zone shows only a marginal (1-2×) rise. Table 4.9 records the pre-flight result for the final model.

Table 4.9: SLG grounding pre-flight check (post-correction)

Zone	Healthy I_a (A)	SLG I_a (A)	Multiplication factor	Status
B2	103.7	7 750.6	74.7×	PASS
B3	137.1	5 737.5	41.9×	PASS
B4	170.6	4 647.7	27.2×	PASS
B5	115.8	6 761.2	58.4×	PASS

Note: All four zones pass. The multiplication factor decreases with electrical distance from T1 (B2 → B4), consistent with the increasing series impedance to each bus; B5 is fed from the smaller T2 and shows a high per-unit rise on its lighter branch.

The physically decreasing multipliers with distance (74.7× at B2 falling to 27.2× at B4) confirm that the fault currents behave according to the feeder impedance gradient rather than a modelling artefact. Only after this gate passes are the 1,000 dataset samples generated, so no SLG sample is produced under an invalid grounding configuration.

4.5 Load-Flow Validation

The healthy-state measurements were cross-checked against analytical calculations. The full-load current of each load centre, computed from its active power at 11 kV, agrees with the simulated healthy current to within about 3% (Table 4.10). The analytical three-phase fault current at Bus B2, computed from the series source, transformer and line impedances, is 7.69 kA, matching the simulated near-bolted fault current of approximately 7.75 kA. These two independent checks confirm that the model reproduces both the normal operating point and the fault-current level correctly.

Table 4.10: Load-flow validation: analytical versus simulated

Bus	P (MW)	Analytical FLC (A)	Simulated healthy I (A)	Deviation
B2	1.950	102.3	103.7	+1.4%
B3	2.600	136.5	137.1	+0.4%
B4	3.250	170.6	170.6	0.0%
B5	2.145	112.6	115.8	+2.8%

Note: Full-load current computed as $I = P/(\sqrt{3}\cdot V_{LL})$ at $V_{LL} = 11$ kV. Agreement within 3% confirms unity-power-factor load modelling and correct in-series measurement placement.

4.6 Random Forest Training

The cost-sensitive Random Forest was trained on the 800-sample training partition using the configuration of Table 4.11. Training a 500-tree forest on 800 samples of 24 features is inherently fast; the reported 43.9 s includes the out-of-bag permutation feature-importance computation. The computationally heavy part of the evaluation is not training but the restoration simulations of Chapter 5, each of which advances the discrete model through 2 s of feeder operation.

Table 4.11: Random Forest hyperparameters

Hyperparameter	Value
Number of trees	500
Features sampled per split	$\lfloor\sqrt{24}\rfloor = 4$
Minimum leaf size	1 (fully grown)
Bootstrap sampling	Enabled (≈ 63.2% per tree)
Misclassification cost	Asymmetric 13×13; fault → Healthy = 12.5×
OOB error and permutation importance	Enabled
Random seed	rng(42), fixed for reproducibility
Training wall-clock time	43.9 s (incl. importance computation)

4.7 Practical Deployment Considerations

Although the present work is simulation-based, practical deployment would require additional hardware, protection-engineering review, and staged commissioning. The proposed logic would not replace primary protection; it would first operate in monitoring mode, then advisory mode, and only later in supervised control mode after sufficient validation. A field installation would introduce a decision-chain latency dominated not by the classifier but by measurement acquisition (one cycle for RMS confirmation), communication and supervisory logic (0.2-2 s), breaker operation (40-100 ms), and tie-switch closing with its interlock checks (1-5 s). The Random Forest evaluation of 24 features is negligible by comparison (< 50 ms). The practical restoration time is therefore estimated at approximately 1-8 s, still far shorter than the 15-60 min typical of manual restoration, but governed by hardware rather than by computation. The present thesis should accordingly be regarded as a feasibility and design study whose results justify further laboratory and hardware-in-the-loop testing, not direct connection to a live mining feeder.

These considerations are consolidated in Table 4.12 as a qualitative risk register. Each entry names a hazard that a field implementation of the scheme would face, the mitigation that is either already embodied in the design or would be required before energisation, and the evidence available from the present study. Several entries are mitigated by results already reported in Chapter 5; several others are mitigated only by work that has not been done, and are marked as such. Setting the two side by side is the point of the register: it separates what this thesis has demonstrated from what a deployment would still have to establish.

Table 4.12: Deployment risk register for a field implementation of the proposed scheme

Ref	Deployment risk	Likelihood × severity	Rating	Mitigation and control
R1	Misclassification isolates the wrong zone	Low × High	Medium	The scheme is supervisory: the conventional IDMT protection of Section 3.5.1 is retained at each bus and operates independently. The zone-to-breaker policy is deterministic and open to review (Section 3.5). The predicted zone was correct in all twelve restoration scenarios (Section 5.6).
R2	Fault classified as Healthy (missed fault)	Low × Very high	High	Cost-sensitive training penalises this error class explicitly (Section 3.4.2, Equation 3.10). No missed faults occurred on the held-out set, nor at any noise level up to 5 % (Tables 5.2 and 5.8). Conventional overcurrent protection remains an independent backstop.
R3	Classifier trained outside the NER-limited earthing regime	High × High	High	Recognised as the principal limitation (Section 6.7) and the highest-priority future work (Section 6.8). Retraining on site data that includes 1-5 A earth faults, and revalidation, are prerequisites for any control action (Appendix C, Stages 1 and 2).
R4	Instrument-transformer error, drift, saturation or channel loss	Medium × Medium	Medium	Accuracy was verified against per-reading Gaussian error up to 5 % (Table 5.8, Figure 5.14). A field scheme additionally requires plausibility, range and completeness checks on the feature vector, which are not implemented here (Section 6.7).

Ref	Deployment risk	Likelihood × severity	Rating	Mitigation and control
R5	Communication latency or loss between the IEDs and the controller	Medium × Medium	Medium	The 0.2-2 s communication allowance is already carried in the 1-8 s practical restoration estimate of this section. A redundant station bus is specified (Appendix C, Table C.1); on loss of data the controller inhibits switching and the feeder reverts to conventional protection.
R6	Cyber-security compromise of the supervisory or control path	Low × Very high	High	Not modelled in this study (Section 6.7). A field scheme requires IEC 62443 zone-and-conduit segregation, authenticated and logged control actions, and no remote path for model update (Appendix C, Table C.2).
R7	Tie-switch closure overloads T2 or backfeeds a faulted zone	Low × High	Medium	The capacity check against the 15 MVA rating of T2 (Equation 3.5) and the voltage-band check (Equation 3.6) are evaluated before the tie is closed, and the terminal-zone lockout held the tie open for every B4 and B5 fault in the twelve scenarios (Section 5.6).
R8	Breaker fails to operate, or status feedback is absent	Low × High	Medium	Breakers are modelled as ideal switches (Section 6.7). A field scheme requires 52a/52b status confirmation, breaker-failure protection and an interlock timeout before restoration is attempted (Appendix C, Table C.2).
R9	Undetected model drift or unauthorised change to the trained model	Medium × Medium	Medium	Training is reproducible from a fixed seed and a versioned archive (Appendix A). Deployment requires change control, a hashed model artefact and revalidation after any retraining (Appendix C, Table C.2).

Ref	Deployment risk	Likelihood × severity	Rating	Mitigation and control
R10	Automation complacency: operators defer to the classifier	Medium × Medium	Medium	The advisory stage displays the predicted class and the proposed switching for operator decision (Section 4.8), and manual override, operator training and event review are retained throughout (Appendix C, Section C.4).

Note: Likelihood and severity are qualitative engineering judgements (Low, Medium, High, Very high) for a first field installation, not probabilities derived from site fault statistics; rating is the resulting exposure on a three-point scale. The register would be re-assessed against recorded event data during the monitoring stage of Appendix C. It addresses deployment of the proposed scheme only and does not replace the site's own protection, earthing or occupational-safety risk assessments.

4.8 Final Topology: Autonomous In-Model Implementation

The two-stage evaluation of Sections 4.4-4.6 established the scheme's behaviour but drove the switching from the MATLAB workspace. To demonstrate that the same scheme can operate autonomously, the validated logic was embedded directly into the Simulink model as an in-model controller, producing the final topology of Figure 4.5. The complete fault-detection, classification, isolation and restoration sequence now runs inside a single continuous simulation.

The controller is realised as a MATLAB Function block (FDIR_Controller) that continuously reads the twenty-four RMS features through a Mux and the simulation time from a Clock. It detects a disturbance from the measurements alone (a current exceeding 1.5 times its healthy baseline, or a bus voltage falling below 0.85 pu), waits 0.5 s for the RMS estimates to settle, and then calls the trained Random Forest to classify the fault. The predicted class drives the deterministic zone-to-breaker policy of Section 3.5, whose five commands are distributed by a Demux to the breaker and tie-switch control inputs, replacing the manual control constants used during dataset generation. The predicted class is shown live on the FDIR_PredictedClass display (0 during healthy operation, and 1-12 once a fault is classified).

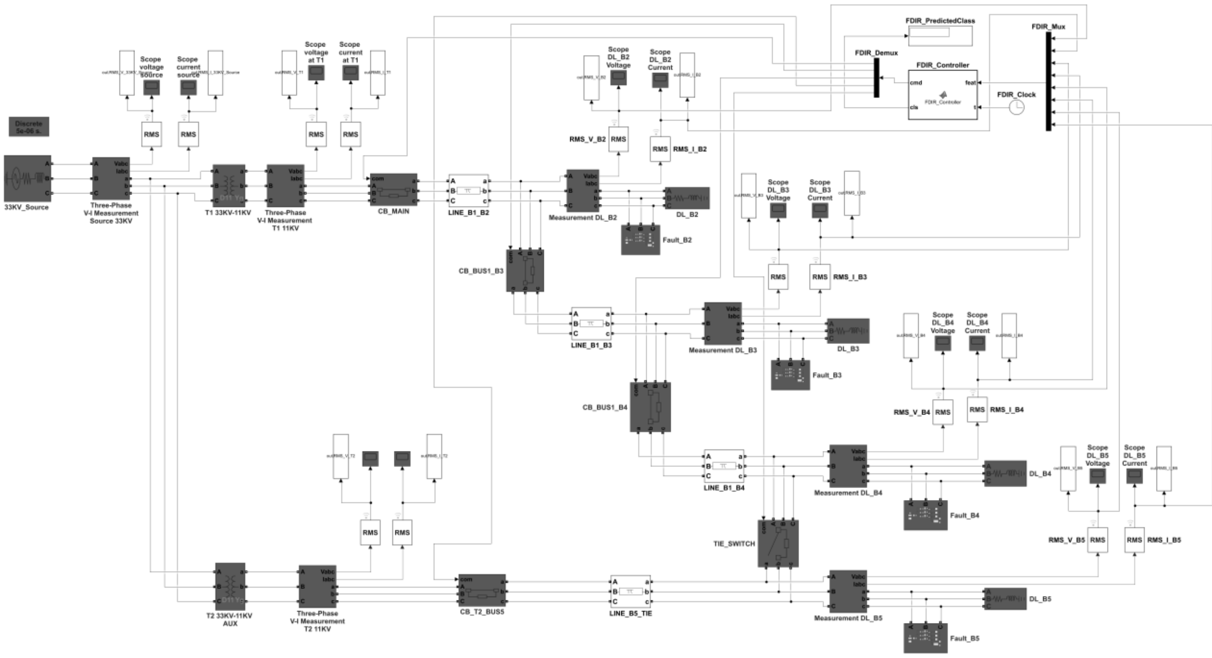

Figure 4.5: Final feeder model with the autonomous in-model FDIR controller. The added blocks (top-right) are the feature Mux, the MATLAB Function controller, the Clock, the command Demux and the predicted-class display; their five outputs drive the breaker and tie-switch control inputs.

With this controller in place, injecting a fault and pressing Run is sufficient: the controller detects the disturbance, identifies the fault type and location from the measurements alone, opens the faulted-zone breakers and, for upstream faults, closes the tie-switch to restore the healthy buses, all without operator intervention. This confirms that the learned-localisation-plus-deterministic-switching architecture can be packaged as a self-contained FDIR automation function rather than a sequence of manual steps. It remains a simulation-based demonstration that uses the same trained model and assumptions as the rest of the study; the in-model capacity check is simplified (the 15 MVA T2 amply covers the feeder load), and hardware-in-the-loop and field validation stay outside the present scope.

4.9 Standards and Design-Reference Conformance

The parameters of the feeder model, the protection benchmark and the acceptance thresholds applied in Chapter 5 are not arbitrary choices; each derives from a recognised standard. Table 4.13 states that mapping explicitly, so that a reader can trace every such value to its source rather than to the author's discretion. The table also records, in its second block, the standards that would govern a field implementation but that this simulation study has not addressed. Listing the second block alongside the first is deliberate: the conformance gap between a simulation and an installation is part of the result, not an omission from it.

Table 4.13: Standards and design references applied in this study

Standard	Aspect governed	Application in this study	Evidence
IEC 60255-151:2009 [18]	Functional requirements for overcurrent and undercurrent protection	The standard-inverse IDMT characteristic and a 0.20 s coordination time interval define the conventional relay benchmark against which the proposed selective isolation is compared.	Section 3.5.1; Table 3.4; Table 5.6
IEC 60909-0:2016 [19]	Calculation of short-circuit currents	The 33 kV supply is represented as a Thevenin equivalent at a 500 MVA three-phase fault level with X/R = 10 at 50 Hz, giving a defensible source impedance rather than an arbitrary one.	Section 3.2.3; Section 4.2.1; Table 4.1
IEEE Std 142-2007 [21]	Earthing of industrial and commercial power systems	Governs the Dyn11 vector groups and the earthing arrangement verified in the SLG pre-flight, and defines the NER-limited earth-fault regime that bounds the validity of the result.	Section 2.2.2; Section 4.2.2; Section 4.4; Section 6.7
IEC 60038:2009 [25]	Standard voltages and supply-voltage tolerance	Supplies the 0.95-1.05 pu band used both as the healthy-state precondition and as the acceptance criterion for post-restoration voltage (Equation 3.6), which is the basis of research question 3.	Section 3.1.2; Section 5.2; Section 5.6; Section 6.4.3
IEC 61869-2 and -3 [26]	Instrument-transformer accuracy classes	Measurement error is injected as a percentage of each RMS reading, matching the way CT and VT accuracy is specified, so that the noise-robustness test is expressed in terms a protection engineer can map onto real instrumentation.	Section 5.7; Table 5.8; Figure 5.14
Applicable to a field implementation, outside the present simulation scope			
IEC 61850 series [20]	Substation communication and interoperability	Not applied. Communications are idealised in the simulation. The standard governs the station bus, reporting and GOOSE signalling that a field scheme would require.	Appendix C, Table C.2; Section 6.7

Standard	Aspect governed	Application in this study	Evidence
IEC 62443 series [27]	Security for industrial automation and control systems	Not applied. No cyber-security modelling was performed. The standard governs the zone-and-conduit segregation and authenticated control of a SCADA-connected scheme.	Appendix C, Table C.2; Table 4.12 (R6)
IEC 60255-1:2009 [28]	Common requirements and type testing for measuring relays	Not applied. No hardware exists to type test. The standard would govern the environmental and type testing of any deployed controller.	Appendix C, Section C.4 (Stage 4)

Note: The first five entries are applied in the present work and can be checked against the sections cited. The remaining three have not been assessed here and are recorded so that the compliance gap between this study and a field installation is explicit. Conformance is claimed only for the specific aspects listed; no certification, type testing or third-party assessment has been carried out against any of these standards.

4.10 Chapter Summary

Chapter 4 confirmed that the methodology was realised in a working MATLAB/Simulink model whose block parameters are documented as editable tables. The healthy state was validated against analytical load flow, the grounding and measurement configuration was verified through a strict SLG pre-flight that all four zones passed, and the cost-sensitive Random Forest was trained under a fixed, reproducible configuration. The project does not simply accept software output; it checks that the simulated electrical behaviour is physically consistent before the data are used for machine learning. The chapter also records the standards from which the model parameters and acceptance criteria are drawn (Table 4.13), and consolidates the hazards that a field implementation would face into a risk register (Table 4.12), so that what the simulation does and does not establish is stated rather than assumed.

Chapter 5 Simulation Results and Analysis

5.1 Introduction

This chapter presents and analyses the simulation results across four dimensions: the electrical behaviour of the feeder under healthy and fault conditions; the classification performance of the Random Forest on the held-out test set; the effectiveness of the closed-loop self-healing restoration across the twelve evaluated scenarios; and a comparative benchmark against conventional protection. All classification metrics are derived from the 200-sample test set withheld before any training activity, and every switching decision reported here was driven by the classifier prediction computed from the pre-switch measurements.

5.2 Healthy-State Electrical Behaviour

Under healthy conditions the feeder satisfies the 0.95-1.05 pu band at every bus, which is the precondition for interpreting the fault and restoration results that follow. Table 5.1 summarises the healthy state, combining the measured phase currents (from the pre-flight of Section 4.4) with the load-flow-estimated per-unit voltages.

Table 5.1: Healthy-state RMS summary

Bus	Process load	Measured I_a (A)	Estimated V (pu)	Status
B2	Dewatering pumps	103.7	≈ 0.986	Within band ✓
B3	Ventilation systems	137.1	≈ 0.981	Within band ✓
B4	Crusher and conveyor	170.6	≈ 0.977	Within band ✓
B5	SX-EW plant (T2)	115.8	≈ 0.983	Within band ✓

Note: Currents are the measured healthy values; per-unit voltages are load-flow estimates from the series source, transformer and line impedances. All buses lie within 0.95-1.05 pu.

5.3 Fault-Condition Electrical Signatures

The classifier separates the fault classes because each fault produces a distinct spatial signature of voltage sag and current rise across the four monitored buses. Figures 5.1 and 5.2 present the per-class voltage and current signatures as heatmaps over the 24 features.

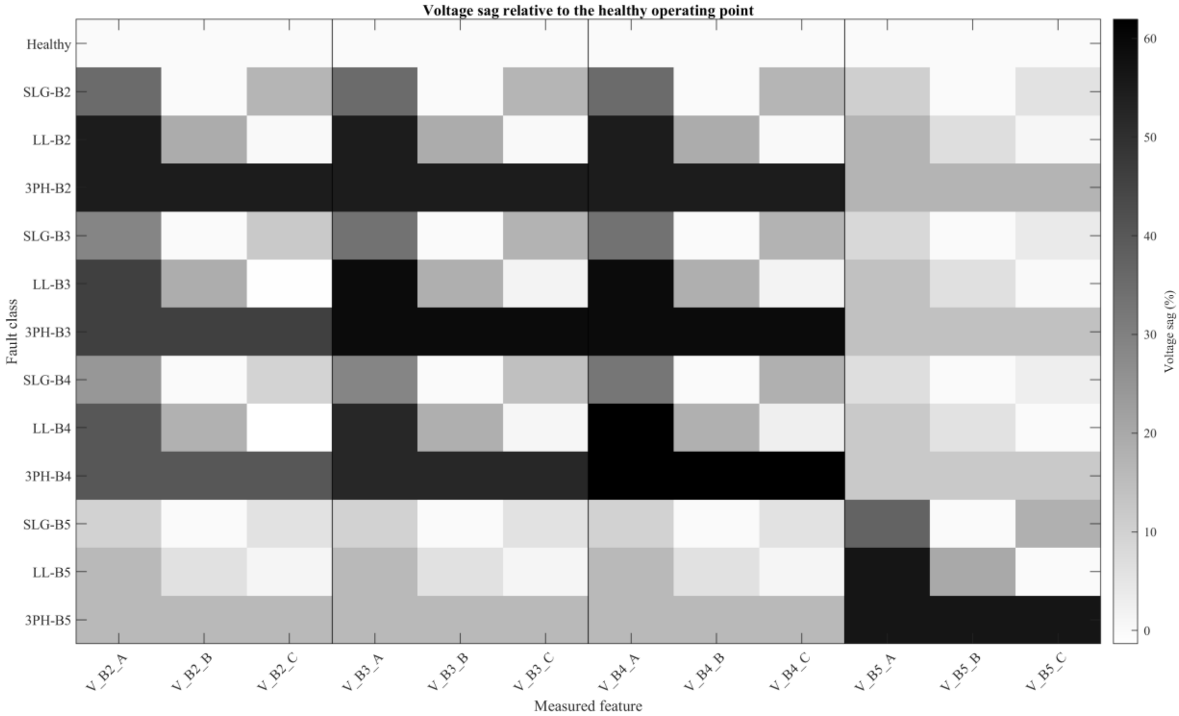

Figure 5.1: Per-class voltage signature (RMS)

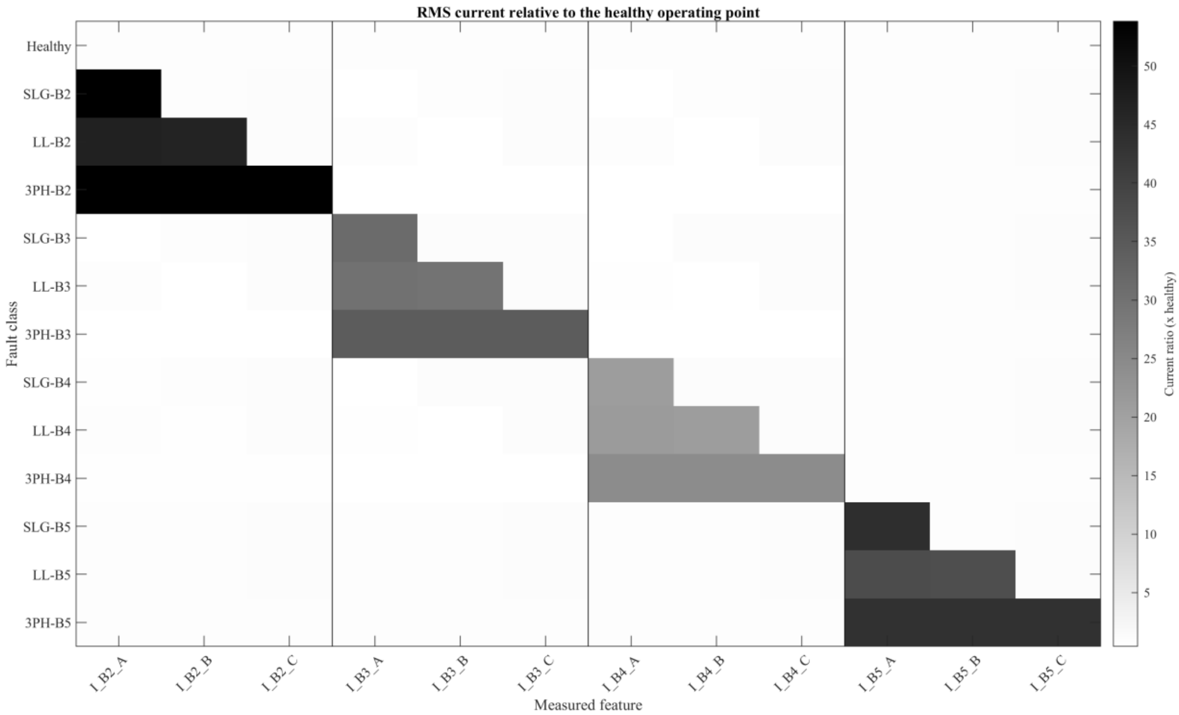

Figure 5.2: Per-class current signature (RMS)

The voltage pattern deepens towards the faulted bus, so a multi-bus feature vector encodes fault location as well as fault presence; three-phase faults produce the largest sags, followed by LL and then SLG. The current signature (Figure 5.2) shows the pronounced rise under near-bolted faults that accounts for the high separability of the simulated classes, and, by contrast, indicates why NER-limited faults, in which this current rise is suppressed by design, would be considerably harder to distinguish.

Figures 5.3-5.6 show four representative live-simulation cases captured during the closed-loop evaluation, spanning both restored and isolated outcomes and both symmetric and asymmetric faults. Each figure shows the Stage-1 fault on the normal network (faulted-bus RMS voltage and current) above the Stage-2 outcome after the prediction-driven switching action.

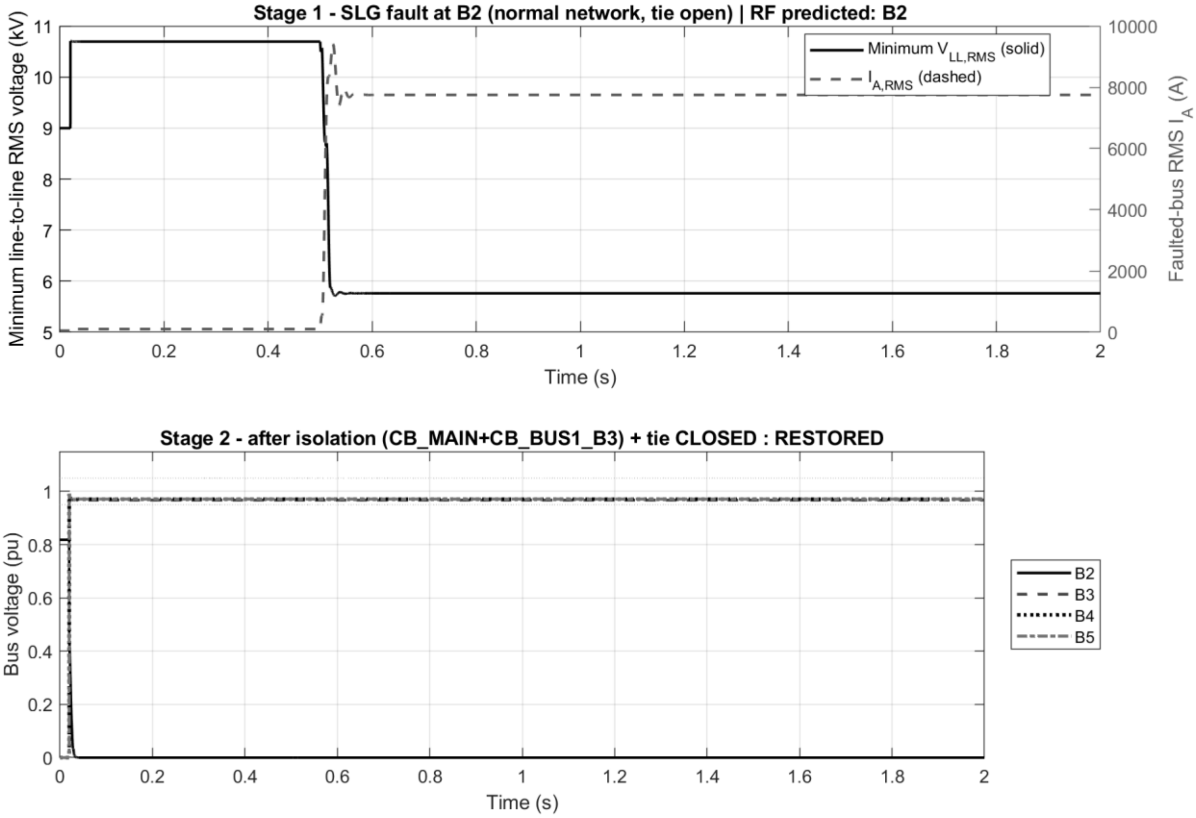

Figure 5.3: SLG fault at B2 - fault and restoration waveforms

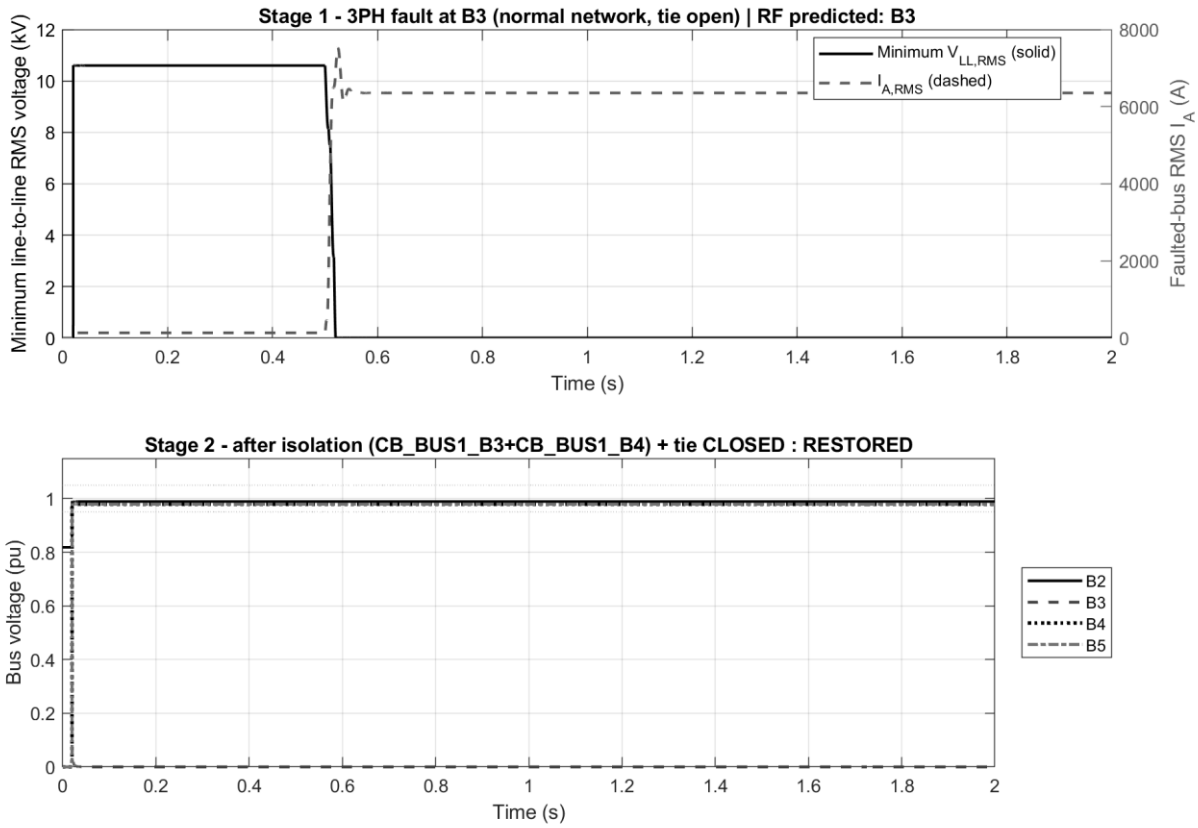

Figure 5.4: Three-phase fault at B3 - fault and restoration waveforms

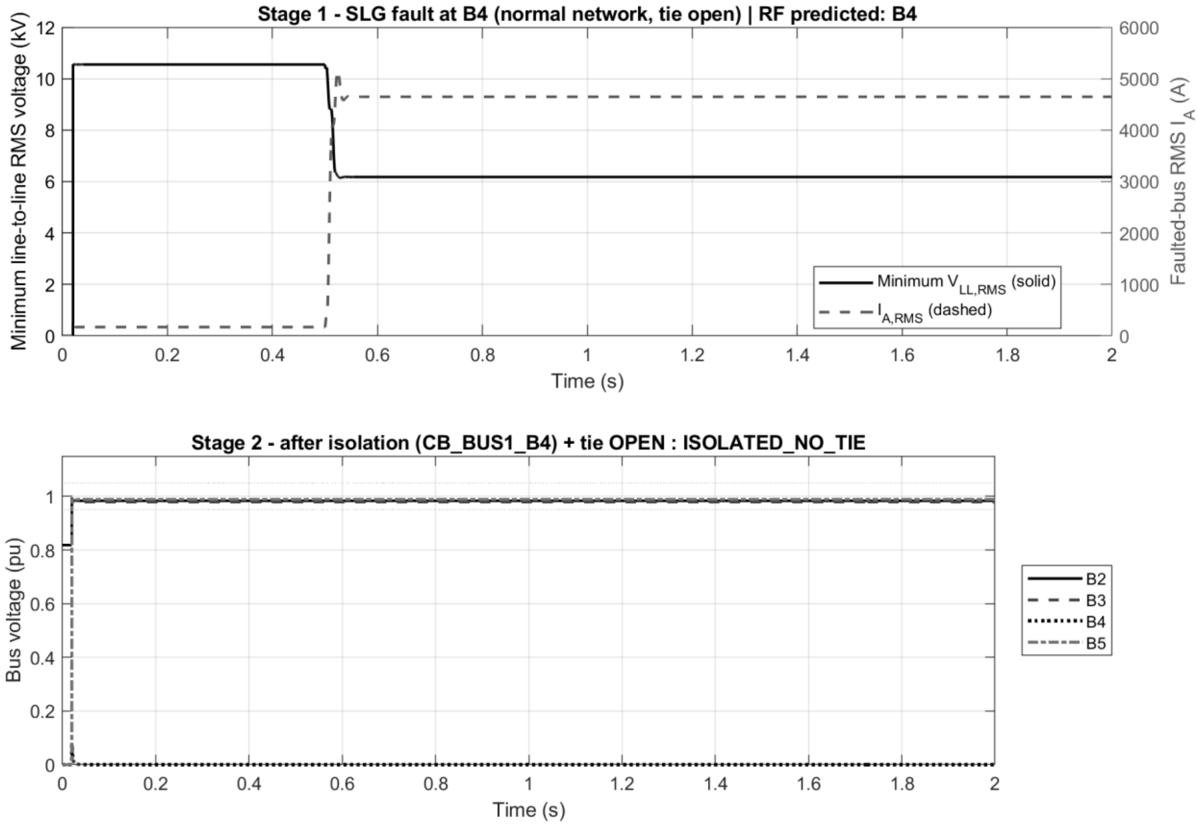

Figure 5.5: SLG fault at B4 - fault and isolation waveforms (tie held open)

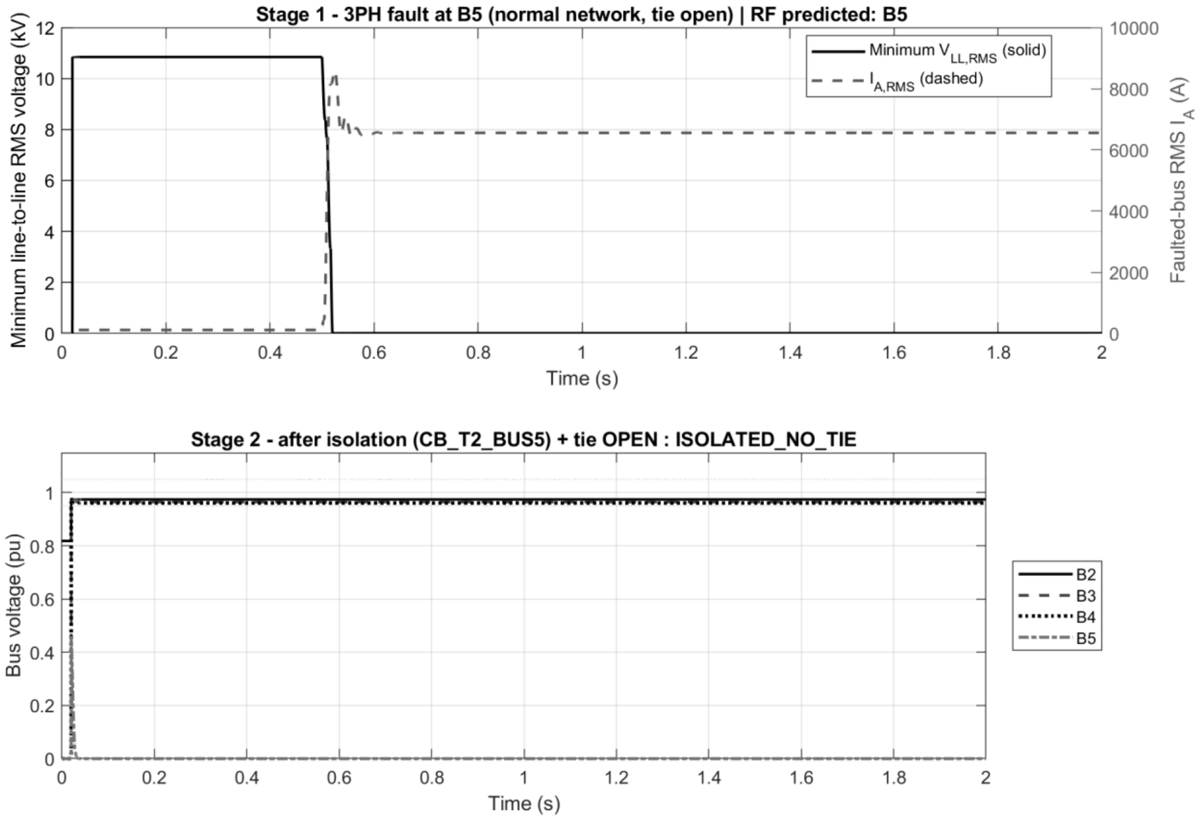

Figure 5.6: Three-phase fault at B5 - fault and isolation waveforms (tie held open)

For the upstream B2 and B3 faults (Figures 5.3-5.4) the classifier predicts the correct zone and the faulted-bus voltage sags sharply during the fault — the minimum line-to-line RMS falling to roughly 5.8 kV for the single line-to-ground case (Figure 5.3) and collapsing toward zero for the three-phase case (Figure 5.4) — after which the faulted bus is de-energised on isolation and the tie closes to restore the downstream healthy buses within the voltage band. For the terminal B4 and B5 faults (Figures 5.5-5.6) the faulted zone is isolated and the tie is correctly held open, because closing it would backfeed the fault; the remaining healthy buses stay on their existing supply. The waveforms confirm that the switching action follows the prediction and that the restoration interlocks behave as designed.

5.4 Machine-Learning Classification Performance

5.4.1 Summary Metrics

Table 5.2 summarises the classifier performance on the held-out test set. The result is bounded by the dataset assumptions, namely ideal measurements and near-bolted fault conditions, and is interpreted accordingly in the next subsection. Figure 5.7 shows that the out-of-bag error stabilises well before 500 trees, justifying the ensemble size.

Table 5.2: Test-set classifier performance summary

Metric	Value	Context
Test-set accuracy	100.00%	200 / 200 correct
Fault detection rate	100.00%	Zero missed faults
Healthy-state detection	100.00%	Zero false alarms

Macro F1-score	1.000	Mean of per-class F1
Five-fold cross-validation	100.00% ± 0.00%	All folds identical
OOB error at N = 500	0.00%	Independent of test set
Training time	43.9 s	500 trees, 800 samples, 24 features

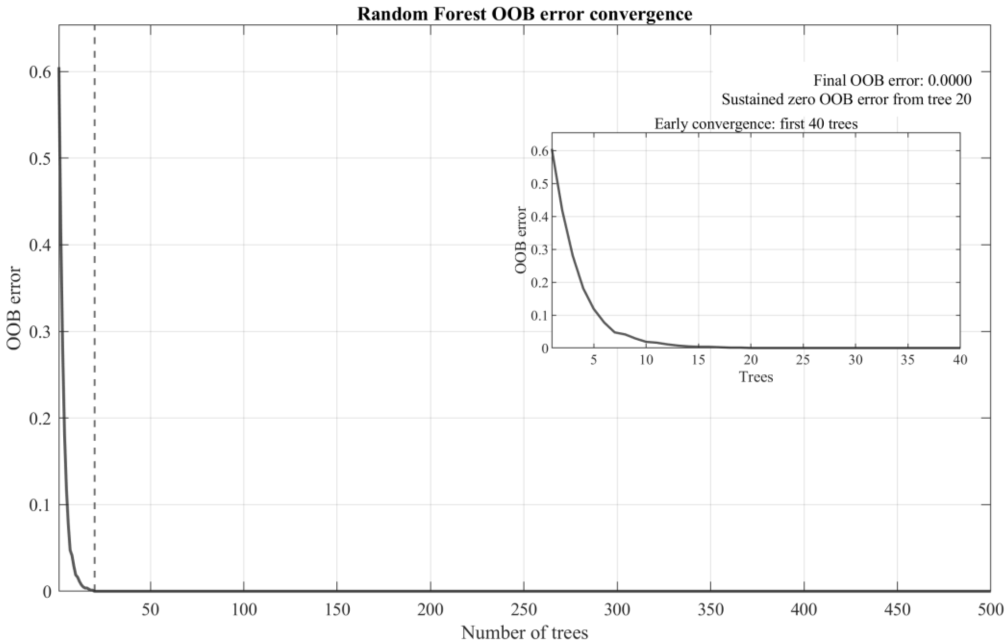

Figure 5.7: Random Forest out-of-bag error convergence

5.4.2 Interpretation of the Zero-Error Result

The zero-error result on the held-out test set is a direct consequence of the controlled simulation environment. After the grounding and measurement configuration was verified, each fault class occupies a clearly separated region of the 24-dimensional RMS feature space; the measurements are ideal, the parameter sweep is structured, and the test set is drawn from the same modelling assumptions as the training set. Under these conditions, complete separation of the classes is the expected outcome rather than a surprising one. The zero out-of-bag error (Figure 5.7) reflects the same separability: because out-of-bag samples are never seen by the trees that evaluate them, it indicates consistent separation within the generated dataset, but it does not rule out model-specific overfitting or guarantee generalisation to unseen feeder conditions.

The result establishes that the chosen feature vector and Random Forest classifier separate the simulated classes under the stated assumptions; it does not establish equivalent performance on a physical feeder, which would introduce measurement noise, CT/VT error, unknown fault impedance, switching transients, communication delay, load variability, and NER-limited earth-fault behaviour. The reported 100% accuracy is therefore a controlled-simulation feasibility result, not a guarantee of field performance. Its engineering significance is narrower but useful: within the studied parameter space, the simulated fault signatures are sufficiently distinct to support reliable selective-isolation decisions. The pertinent question is not why the accuracy reaches 100%, but which assumptions produce that figure, and which future tests would challenge it, a point revisited in the comparative benchmark and in Chapter 6.

5.4.3 Confusion Matrix and Per-Class Metrics

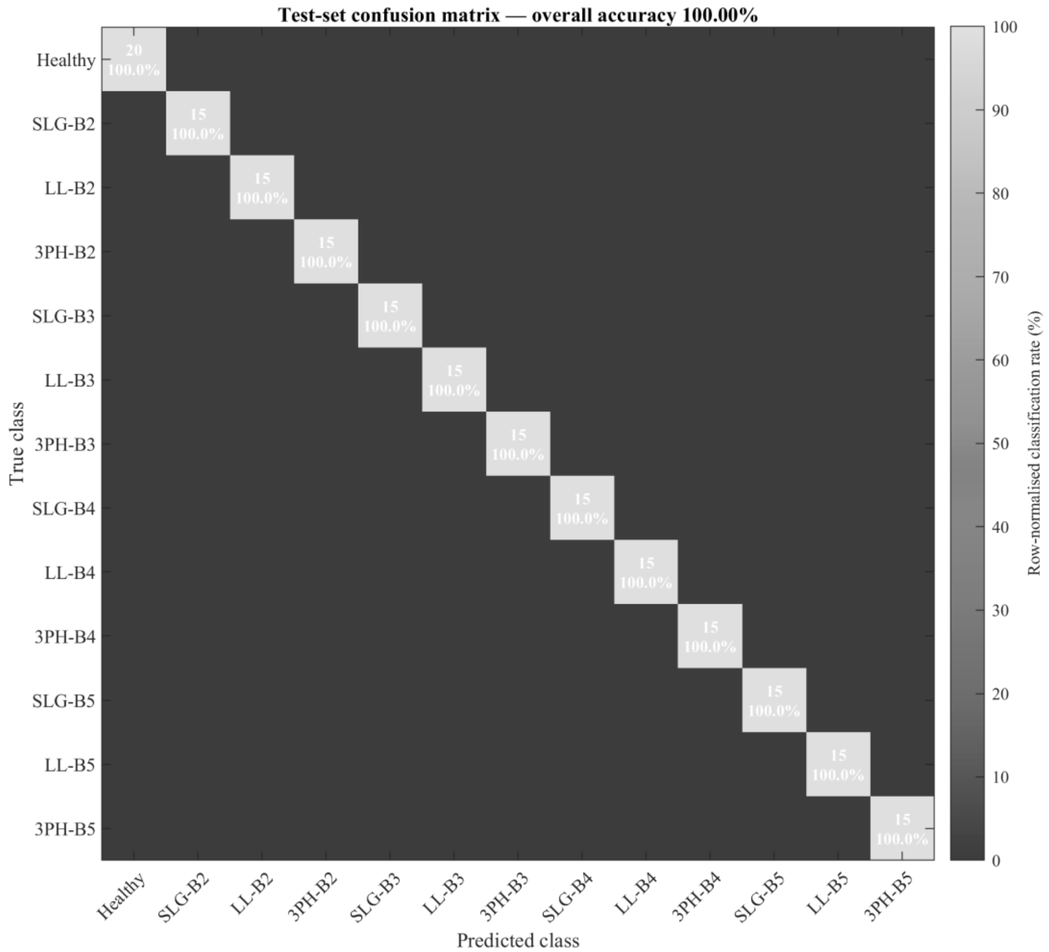

Figure 5.8: Test-set confusion matrix

Figure 5.8 resolves performance class by class, with no off-diagonal entries for this controlled dataset, confirming that the overall accuracy is not masking confusion between specific fault classes. Table 5.3 and Figure 5.9 report the per-class precision, recall and F1-score, all equal to 1.000, confirming that performance is uniform across classes and not sustained by the Healthy class or a subset of easily separable faults.

Table 5.3: Per-class classifier metrics

Class	Label	Precision	Recall	F1	n
0	Healthy	1.000	1.000	1.000	20
1	SLG-B2	1.000	1.000	1.000	15
2	LL-B2	1.000	1.000	1.000	15
3	3PH-B2	1.000	1.000	1.000	15
4	SLG-B3	1.000	1.000	1.000	15
5	LL-B3	1.000	1.000	1.000	15
6	3PH-B3	1.000	1.000	1.000	15
7	SLG-B4	1.000	1.000	1.000	15
8	LL-B4	1.000	1.000	1.000	15
9	3PH-B4	1.000	1.000	1.000	15

10	SLG-B5	1.000	1.000	1.000	15
11	LL-B5	1.000	1.000	1.000	15
12	3PH-B5	1.000	1.000	1.000	15
Macro	arith. mean	1.000	1.000	1.000	200

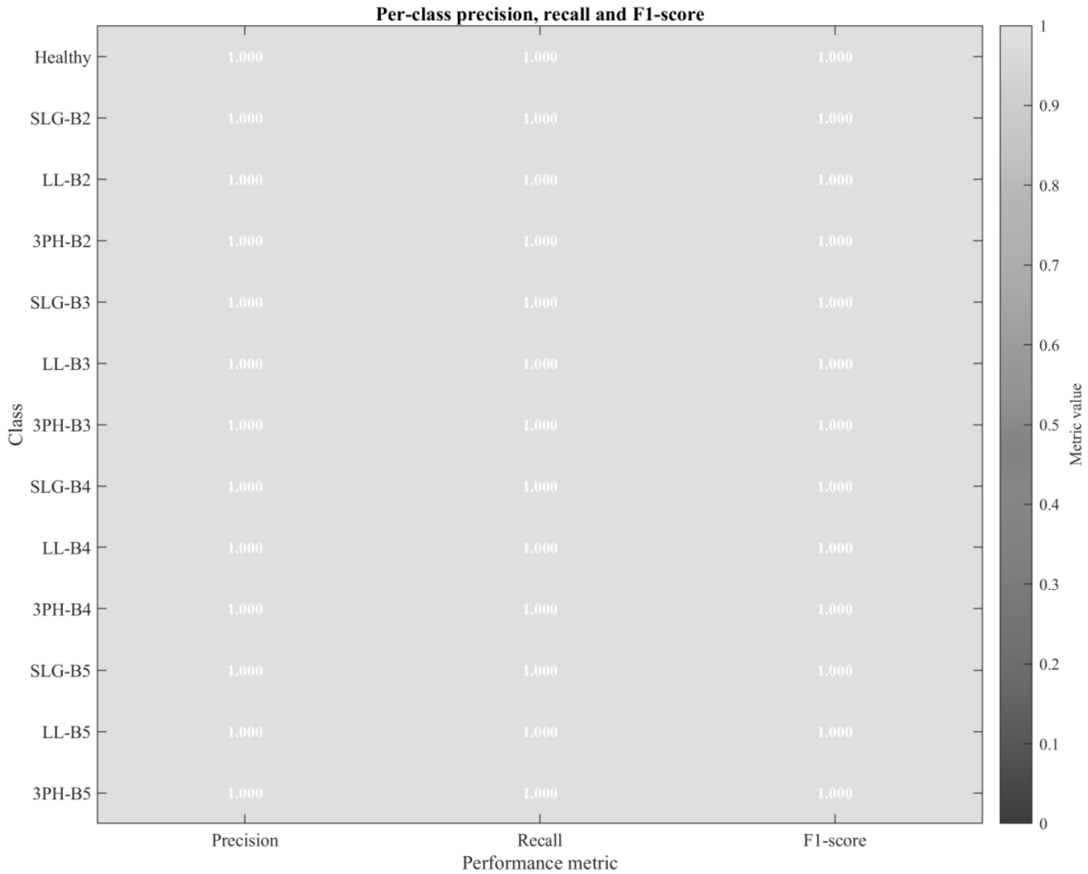

Figure 5.9: Per-class precision, recall and F1-score

With 15 test samples per fault class, the per-class estimates are necessarily coarse: a single misclassification would change a class recall by 6.7 percentage points. The uniform 1.000 values should therefore be read as consistency within a small controlled test set rather than as tight statistical bounds, a caveat carried forward into the limitations.

5.4.4 Cross-Validation and Feature Importance

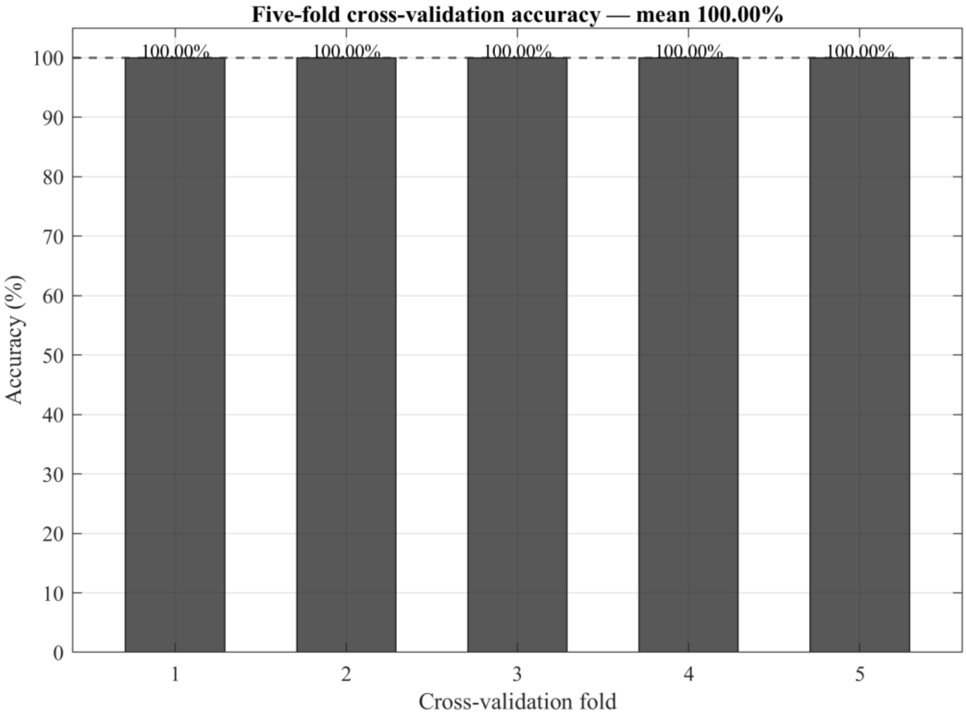

Figure 5.10: Five-fold cross-validation accuracy

Five-fold cross-validation (Figure 5.10) returns identical accuracy across all folds, reducing the likelihood that the result depends on a single favourable 80/20 split, although it does not substitute for field validation. Figure 5.11 ranks the features by out-of-bag permutation importance, relating the classifier decision to specific bus measurements and confirming that the distributed voltage and current channels each contribute to localisation.

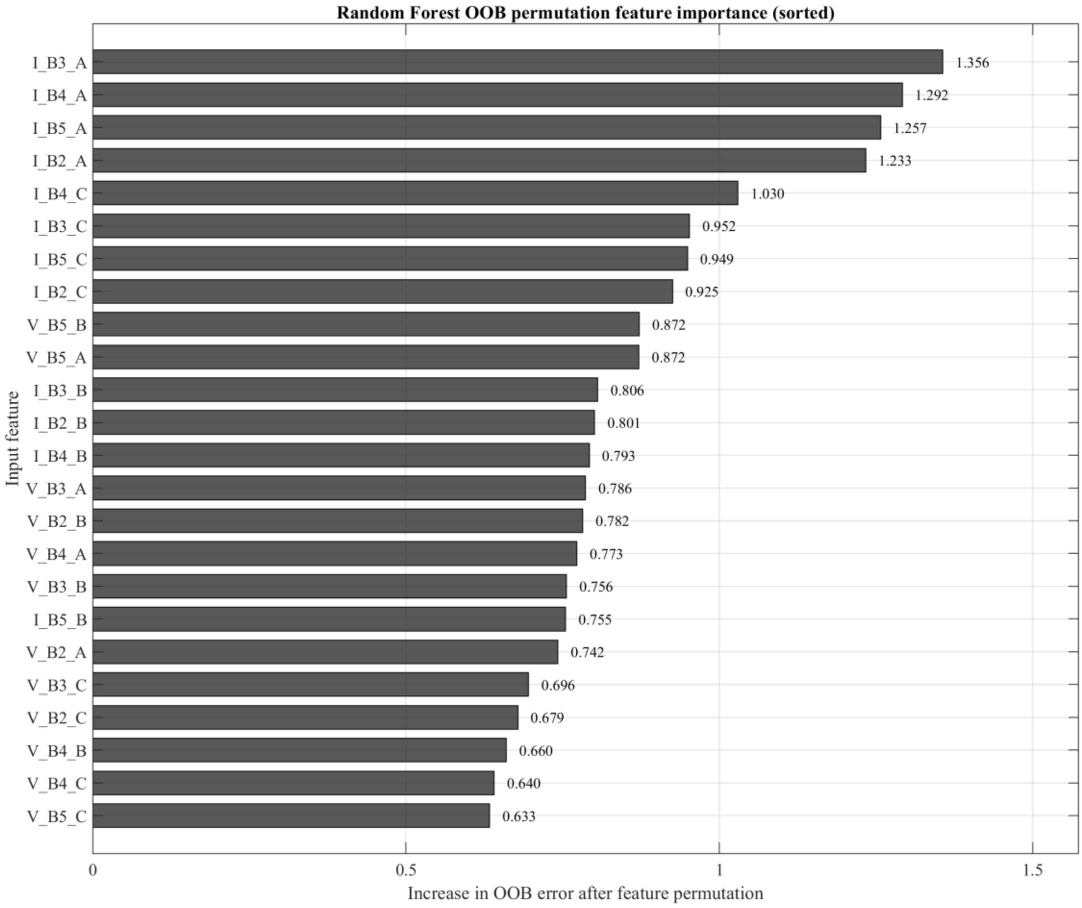

Figure 5.11: OOB permutation feature importance

5.5 Selective Isolation Performance

In this series-radial topology, the breaker to be tripped is determined uniquely by the predicted fault-bus location. Selective isolation accuracy, the proportion of fault test samples whose predicted location maps to the correct breaker command, is therefore a restatement of the location component of the classifier accuracy. Since all 180 fault test samples are correctly classified, the isolation logic selects the correct breaker in every case, and all 20 Healthy samples produce zero false isolation commands.

5.6 Self-Healing Restoration Performance

The restoration suite evaluated twelve scenarios, the four fault zones at three loading levels (0.70, 1.00, 1.30 pu), each driven end-to-end by the classifier prediction. Table 5.4 lists the outcomes. In all twelve cases the predicted zone was correct and the faulted zone was isolated; the six upstream cases (B2 and B3) closed the tie to restore the downstream healthy buses, while the six terminal cases (B4 and B5) correctly held the tie open. Figures 5.12 and 5.13 summarise the decision outcomes and the post-restoration voltages.

Table 5.4: Closed-loop restoration results (twelve scenarios)

Zone	LM	Pred (OK)	Status	Isolated	Tie	Restored / on T1	V_{post} (pu)
B2	0.70	B2 ✓	RESTORED	B2	Closed	B3+B4	0.972 / 0.974
B2	1.00	B2 ✓	RESTORED	B2	Closed	B3+B4	0.967 / 0.971
B2	1.30	B2 ✓	RESTORED	B2	Closed	B3+B4	0.963 / 0.967
B3	0.70	B3 ✓	RESTORED	B3	Closed	B4 (B2 on T1)	0.981
B3	1.00	B3 ✓	RESTORED	B3	Closed	B4 (B2 on T1)	0.979
B3	1.30	B3 ✓	RESTORED	B3	Closed	B4 (B2 on T1)	0.976
B4	0.70	B4 ✓	ISOLATED_NO_TIE	B4	Open	B2+B3 on T1	—
B4	1.00	B4 ✓	ISOLATED_NO_TIE	B4	Open	B2+B3 on T1	—
B4	1.30	B4 ✓	ISOLATED_NO_TIE	B4	Open	B2+B3 on T1	—
B5	0.70	B5 ✓	ISOLATED_NO_TIE	B5	Open	B2+B3+B4 on T1	—
B5	1.00	B5 ✓	ISOLATED_NO_TIE	B5	Open	B2+B3+B4 on T1	—
B5	1.30	B5 ✓	ISOLATED_NO_TIE	B5	Open	B2+B3+B4 on T1	—

Note: Pred (OK) = Random Forest predicted zone and whether it matched the injected zone; every case matched. For terminal-zone (B4, B5) faults the tie is held open by design, so no restoration voltage is reported. Post-restoration voltages for the six RESTORED cases span 0.963-0.981 pu.

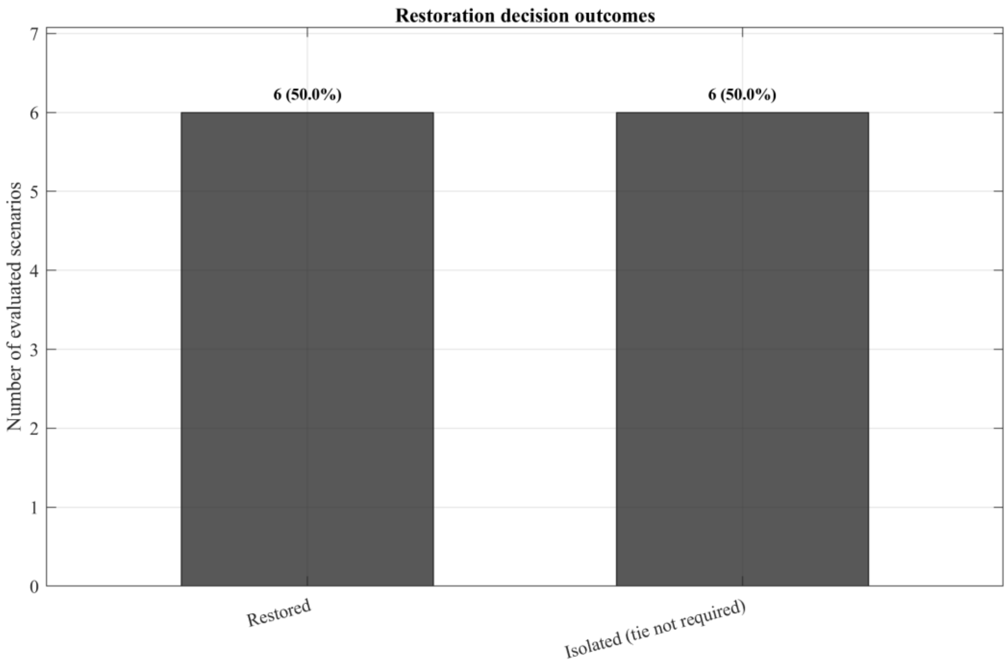

Figure 5.12: Restoration decision outcomes by status

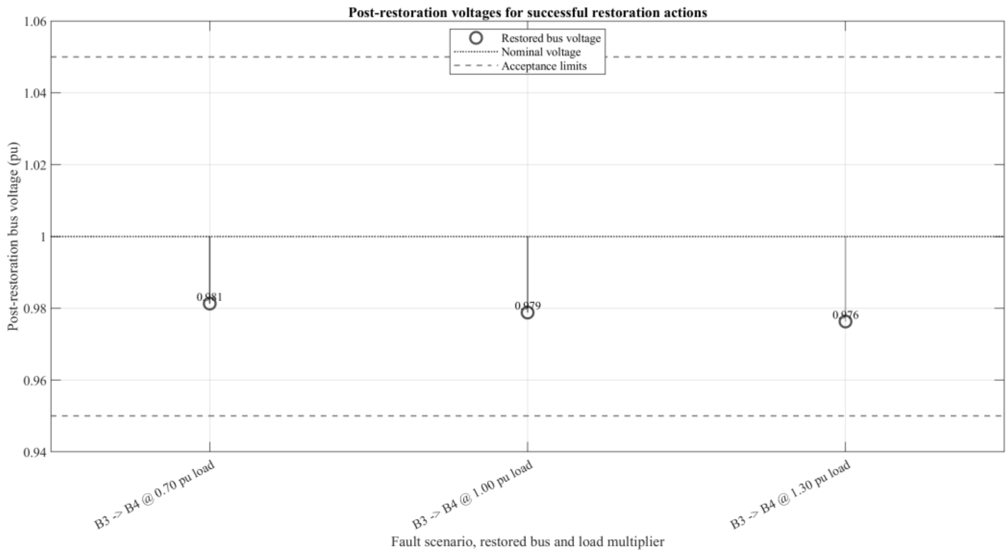

Figure 5.13: Post-restoration voltage - eligible (RESTORED) cases

The post-restoration voltages (Figure 5.13) remain within the 0.95-1.05 pu band in every eligible case, contracting from 0.981 pu at light load to 0.963 pu at heavy load, a physically expected trend that justifies the capacity and voltage checks performed before the tie is closed. The reconnected apparent power ranged from 3.9 MVA to 9.9 MVA, all within the 15 MVA T2 rating, so the capacity constraint of Equation (3.5) was satisfied throughout.

5.7 Comparative Benchmarking

The proposed scheme is benchmarked against a conventional protection response on the same feeder, fault set, and operating conditions. The conventional benchmark is assumed to trip the nearest upstream breaker that clears the fault and, lacking automated restoration, to leave all healthy sections downstream of that breaker de-energised until a crew performs manual switching. This is an analytical/computational benchmark computed from the saved results; the conventional scheme was not physically tested. The primary service-continuity measure is

$N_{healthy,off}$, the number of healthy zones left de-energised after isolation, computed for each fault location from the restoration results and shown in Figure 6.1. Table 5.5 first records why the perfect classification result must be read as feasibility evidence, and Table 5.6 then sets out the head-to-head comparison.

Table 5.5: Perfect-accuracy limitation check

Factor	Condition in this thesis	Expected field effect
Measurement quality	Ideal RMS signals from Simulink	Noise, calibration error and CT/VT saturation can blur class boundaries.
Fault impedance	0.001-5.0 Ω controlled range	Higher/variable impedance reduces current rise and can overlap classes.
Earthing practice	NER-limited faults not modelled	Earth-fault current may be limited to a few amperes, weakening discrimination.
Operating variability	Selected LM and onset times	Real mines include motor starts, process changes and unbalanced loading.
Validation source	Synthetic train/test from one model	Independent relay records or HIL tests are needed before deployment claims.

Table 5.6: Comparative protection benchmark

Criterion	Conventional wide-trip scheme	Proposed ML-assisted scheme	Observed improvement	Engineering interpretation
Faulted zones isolated	1 (fault cleared)	1 (fault cleared)	Equal	Both clear the fault; safety is preserved.
Healthy zones de-energised (B2/B3/B4/B5 faults)	2 / 1 / 0 / 0	0 / 0 / 0 / 0	Up to 2 zones saved	Prediction-driven sectionalising plus tie restoration keeps healthy load in service.
Fault localisation	None; crew investigation	Explicit bus-level prediction	Qualitative gain	Enables selective action instead of wide tripping.
Healthy-load restoration	Manual, 15-60 min	Automatic for eligible zones (6/12 cases)	Minutes → seconds	Tie closes only when capacity and voltage checks pass.
Post-restoration voltage	Not applicable	0.963-0.981 pu (within band)	Verified acceptable	Voltage stays in the 0.95-1.05 pu band under all tested loads.
Tie-switch security	Not applicable	Tie open for all B4/B5 faults (6/6)	No backfeed	The tie is never closed into an un-cleared fault.
Implementation complexity	Low; proven	Higher; needs sensing, comms, validation	Trade-off	Added capability requires measurement and

				communication infrastructure.

Note: Counts are computed from the saved restoration results at LM = 1.00. The comparison applies to the modelled feeder, fault set and operating conditions only; it does not imply universal superiority, and conventional protection remains the certified, fail-safe primary layer.

The benchmark shows the proposed method to be strongest precisely where conventional protection is weakest: locating the faulted section and restoring healthy load where the topology permits. Across the four fault locations the conventional response leaves up to two healthy zones de-energised, whereas the proposed scheme leaves none, because it isolates only the faulted zone and restores the eligible downstream buses through the tie. Conventional protection nonetheless remains the indispensable safety layer because it is certified and fail-safe, so the practical direction is supervised integration rather than replacement.

A first robustness check addresses whether the cost matrix, rather than class separability, produced the zero missed-fault result. A standard Random Forest was retrained without the cost matrix on the identical training split, with the same bootstrap seed, so that the only difference from the cost-sensitive model is the cost argument. As Table 5.7 shows, the two configurations are indistinguishable: both achieve 100% test accuracy with zero missed faults and zero false alarms. On this highly separable dataset the cost matrix therefore does not change the missed-fault outcome; it is a safety-oriented design choice whose benefit would emerge only for a harder, less separable problem such as NER-limited earth faults.

Table 5.7: Cost-sensitivity ablation (standard versus cost-sensitive Random Forest)

Configuration	Test accuracy	Fault recall	Missed faults	False alarms
Standard RF (no cost)	100.00%	100%	0	0
Cost-sensitive RF (12.5×)	100.00%	100%	0	0

Note: same 200-sample held-out test set as the main result (rng 42). Both forests were trained on the identical 800-sample training partition with the same bootstrap seed, differing only in the 13×13 cost matrix. Source: ablation_cost_sensitivity_v2.txt.

A further robustness check quantifies the effect of realistic measurement error. Multiplicative Gaussian noise, with standard deviation equal to a stated percentage of each RMS reading (matching how instrument-transformer accuracy classes are specified), was added to the held-out test features and the trained model re-evaluated over 100 Monte-Carlo repetitions per level (Table 5.8, Figure 5.14). The classifier degrades gracefully: mean accuracy stays above 98% up to a 2% per-reading error and is still 94.6% (worst case 91.0%) at a severe 10% error. Critically for protection, the fault-detection rate remains 100% with zero missed faults at every noise level, so the operationally important behaviour, never mistaking a fault for the healthy state, is preserved even under substantial measurement error.

Table 5.8: Classifier robustness to measurement noise

Measurement error σ (%)	Mean accuracy	Worst-case accuracy	Fault recall	Missed faults
0	100.00%	100.00%	100%	0
0.5	100.00%	100.00%	100%	0

1	99.97%	99.50%	100%	0
2	99.84%	98.50%	100%	0
5	97.99%	95.00%	100%	0
10	94.56%	91.00%	100%	0

Note: Multiplicative per-reading Gaussian error added to the 200-sample test set; 100 Monte-Carlo repetitions per level. Source: noise_robustness_v2.txt.

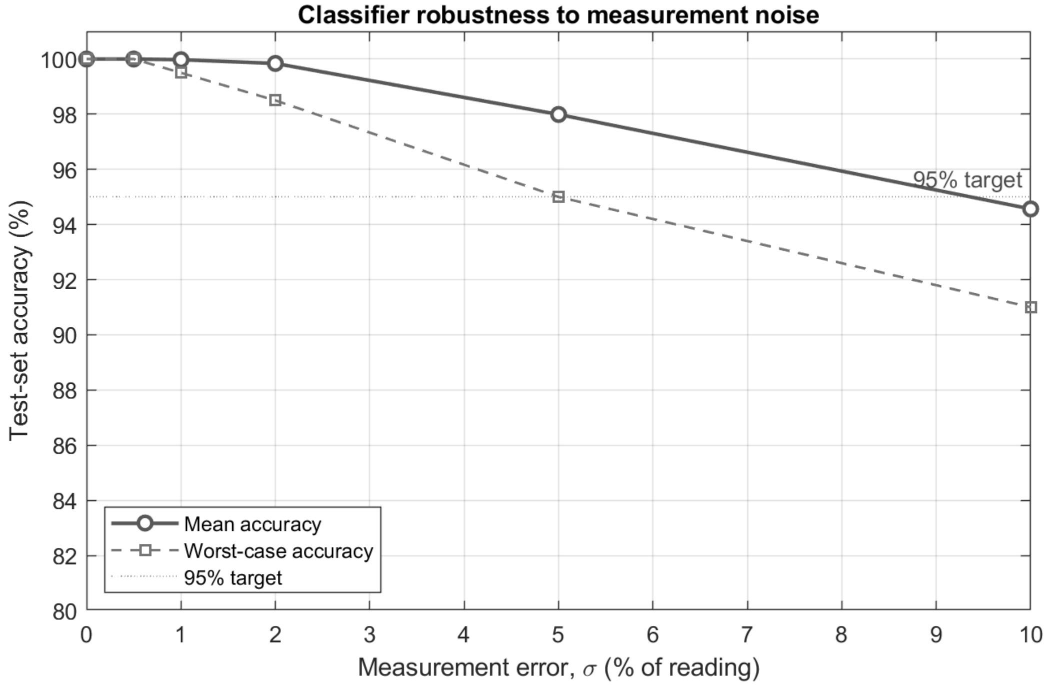

Figure 5.14: Classifier accuracy versus measurement noise

Finally, generalisation to conditions between the training-grid points was tested. Seventy-two off-grid cases were simulated at fault resistances (0.05, 0.30, 2.50 Ω), load multipliers (0.78, 1.20 pu) and an onset time (0.625 s) lying strictly between the sampled grid values, and classified with the existing model without retraining (Table 5.9). All 72 cases were classified correctly, giving 100% exact-class and zone-localisation accuracy with zero missed faults across all three fault types. This indicates the class separation is a property of the parameter region rather than only of the sampled points, although it remains bounded by the near-bolted, ideal-sensor assumptions and does not extend to the NER-limited regime.

Table 5.9: Off-grid (interpolation) generalisation test

Metric	Off-grid result
Off-grid cases (Rf, LM, ton between grid points)	72
Exact-class accuracy (type + location)	100.00% (72/72)
Zone-localisation accuracy	100.00% (72/72)
Missed faults (fault predicted Healthy)	0
SLG / LL / 3PH exact accuracy	100% / 100% / 100%

Note: Fault resistances, load multipliers and onset time lie strictly between the training-grid points; predictions use the existing rf_model_v2.mat. Source: interpolation_test_v2.txt.

A final analysis addresses whether the Random Forest is necessary at all, given how strongly the per-zone current signatures separate the fault classes. Two much simpler alternatives were evaluated on the same 200-sample held-out test set, with no re-simulation: a max current-ratio zone detector, which localises the fault to the bus with the largest ratio of measured to healthy current (an argmax over zones, with a healthy threshold), and a single unpruned decision tree solving the full thirteen-class problem. Table 5.10 reports the outcome.

Table 5.10: Comparison with rule-based baselines

Method	Zone-localisation accuracy	Full 13-class accuracy	Interpretability
Max current-ratio rule	99.5%	— (localisation only)	High (one threshold)
Single decision tree	100%	100%	High (12 splits)
Random Forest (this work)	100%	100%	Moderate (500 trees)

Note: same 200-sample held-out test set as the main result (rng 42), computed on the existing dataset with no additional simulation. The current-ratio rule localises the faulted zone only and does not classify fault type, so its full-class entry is not applicable.

On this highly separable dataset the zone localisation that drives the switching action is recovered almost perfectly by a trivial current-ratio rule (99.5%), and the full thirteen-class classification is matched exactly by a single twelve-split decision tree (100%). The Random Forest is therefore not strictly necessary for the present, idealised problem; its ensemble averaging earns its place through robustness rather than raw accuracy, degrading gracefully under measurement noise and generalising to off-grid conditions (both shown above), advantages that would become decisive only in the harder, less separable regimes — NER-limited earth faults, or noisy and partially missing measurements — identified as future work. Reporting this baseline explicitly guards against overstating the role of the learned model on a clean synthetic dataset.

5.8 Autonomous Operation

Section 4.8 embedded the scheme into the model as an autonomous controller. Figure 5.15 shows the result of running that model with a single line-to-ground fault applied at Bus B2 at t = 0.5 s, captured over one continuous simulation. Before the fault, all buses sit within the 0.95-1.05 pu band. At t = 0.5 s the fault drives Buses B2, B3 and B4 down to about 0.69 pu, because they share the T1 feeder, while B5 sags to about 0.89 pu. The controller detects the disturbance, waits for the RMS estimates to settle, and at t = 1.001 s classifies the fault as SLG-B2 from the measurements alone; it then opens CB_MAIN and CB_BUS1_B3 and closes the tie-switch. Bus B2 collapses to zero, isolated and de-energised, while Buses B3, B4 and B5 recover to about 0.97 pu, back inside the acceptance band, within a cycle of the switching action.

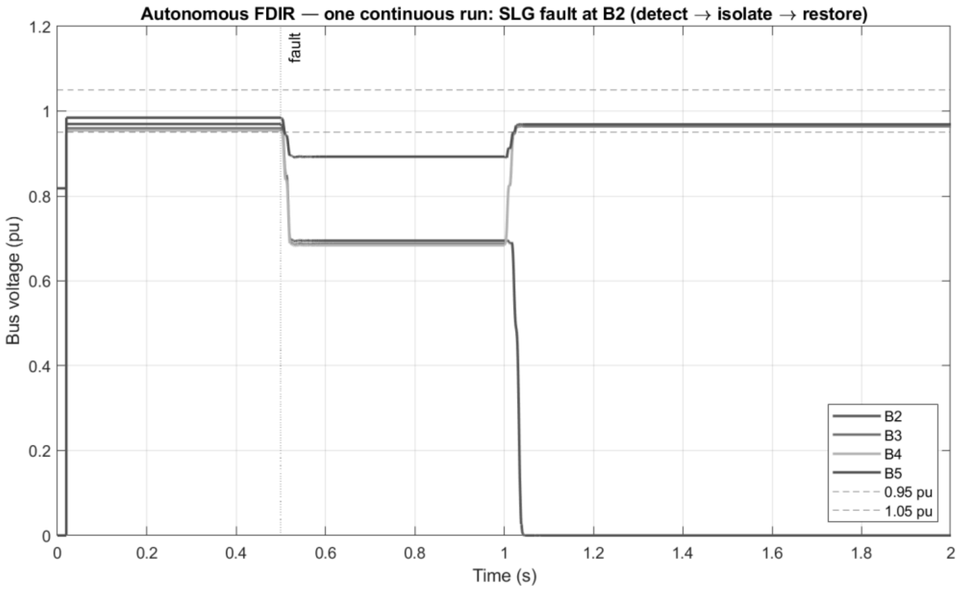

Figure 5.15: Autonomous FDIR operation in a single continuous simulation. An SLG fault at B2 (t = 0.5 s) is detected, classified as SLG-B2 at t = 1.001 s, isolated (B2 to 0 pu), and the healthy buses restored through the tie (~0.97 pu) by t ~ 1.03 s.

The entire detection, classification, isolation and restoration sequence therefore executes in a single simulation with no operator intervention, reproducing the two-stage results of Sections 5.4-5.6 as one continuous event. As throughout the study, this is a controlled-simulation demonstration using the trained model of Chapter 4; it establishes that the scheme can act autonomously within the model, not that it is ready for field deployment.

5.9 Chapter Summary

Chapter 5 showed that the simulated feeder behaves correctly under healthy and faulted conditions, that the Random Forest separates the controlled dataset without error, that the closed-loop restoration isolates the faulted zone and restores eligible buses inside the voltage band while never backfeeding a terminal fault, and that the scheme improves service continuity relative to a conventional wide-trip response on this feeder. The chapter also explained why the perfect-accuracy result occurs and what would reduce it in a real feeder. The key engineering outcome is that the concept works in the simulated design space and is strong enough to justify a next validation stage.

Chapter 6 Conclusions and Recommendations

6.1 Introduction

This chapter draws together the findings of the research, assesses achievement against each specific objective, answers the three research questions, acknowledges the principal limitations of the simulation-based study, and recommends priorities for future work.

6.2 Summary of Research

This thesis has designed and evaluated, by MATLAB/Simulink simulation, a machine-learning-assisted fault-isolation and service-restoration scheme for a representative 33/11 kV mining distribution feeder. It is presented as a proof-of-concept study: all data are synthetic, and hardware-in-the-loop and field validation lay outside the approved project scope.

6.3 Achievement of Research Objectives

Table 6.1 maps each approved objective to the part of the thesis that addresses it, substantiating the assessment of achievement rather than asserting success in general terms.

Table 6.1: Achievement of research objectives

No.	Objective	Achievement
1	Model a representative five-bus 33/11 kV mining feeder in MATLAB/Simulink.	Feeder model built with two transformers, four load centres, breakers, fault blocks, in-series measurements, and a normally-open tie-switch.
2	Generate simulated fault data under controlled conditions.	A 1,000-sample, 13-class dataset generated from healthy, SLG, LL and 3PH faults across four buses and a bounded parameter sweep.
3	Train and evaluate a Random Forest classifier using RMS V-I features.	Cost-sensitive 500-tree forest separated the classes with zero test-set error; limitations explicitly identified.
4	Assess isolation and restoration logic under voltage constraints.	Prediction-driven switching isolated the faulted zone in all cases and restored eligible buses within 0.95-1.05 pu.

6.4 Answers to the Research Questions

6.4.1 Question 1 - Classification of simulated feeder states

Within the controlled simulation dataset, the answer is affirmative. The Random Forest separated the 13 system states with zero test-set error, a result best read as evidence that the selected RMS feature vector is informative under the modelled near-bolted conditions rather than as proof of field performance.

6.4.2 Question 2 - Effect of cost-sensitive classification

Within the simulated dataset, the cost-sensitive formulation produced no missed faults among the held-out fault samples. An ablation confirms this is not attributable to the cost matrix: a standard Random Forest trained without the cost matrix on the identical split gives the same outcome — 100% accuracy and zero missed faults (Table 5.7). The cost matrix is therefore a safety-oriented design choice rather than the cause of the zero-missed-fault result, which instead reflects the high separability of the corrected near-bolted classes; its benefit would emerge only for a harder, less separable problem.

6.4.3 Question 3 - Post-restoration voltage within the band

For the restoration-eligible cases evaluated in the model, the answer is affirmative: post-restoration voltages remained between 0.963 pu and 0.981 pu across the tested loading conditions, all within the 0.95-1.05 pu band. The result applies to the eligible restored section of the single-tie-switch topology and does not extend to full restoration of all healthy buses following upstream faults.

6.5 Scenario Comparison

Figure 6.1 expresses the technical result as an operational outcome, quantifying how selective isolation and restoration reduce the number of healthy zones left de-energised. Figure 6.2 summarises the computed performance of the scheme against the study criteria, and Table 6.2 contrasts the proposed logic with conventional manual restoration for a representative fault.

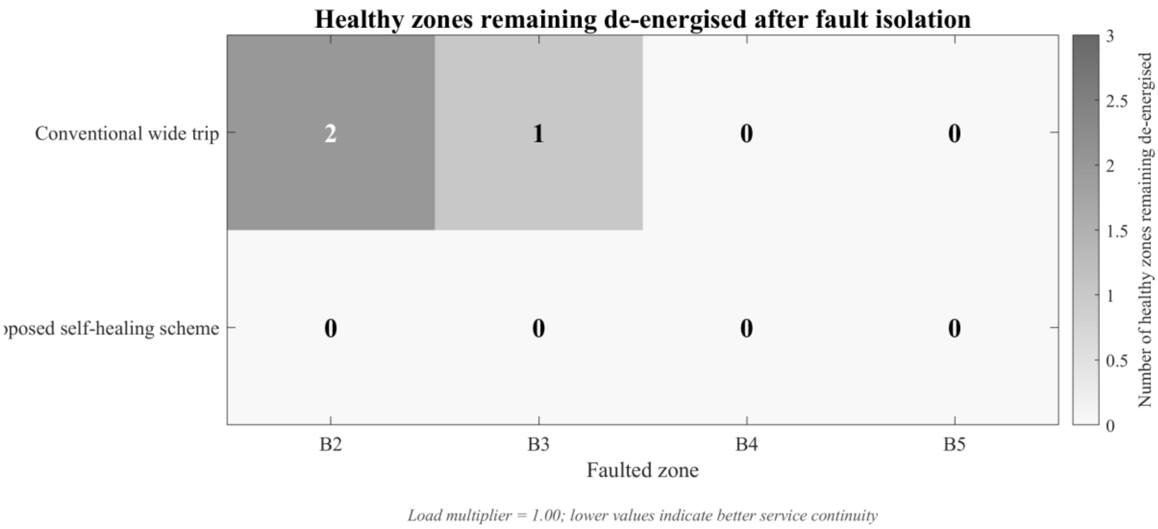

Figure 6.1: Healthy zones remaining de-energised after fault isolation

Figure 6.2: Computed performance summary of the proposed scheme

Table 6.2: Bus-fault scenario comparison (conventional versus proposed)

Performance dimension	Conventional wide-trip	Proposed ML self-healing
Healthy zones de-energised (B2 fault)	2 (B3, B4)	0 (B3, B4 restored via tie)
Fault bus identified	No	Yes (bus-level prediction)
Restoration of healthy buses	Manual, 15-60 min	Automatic where eligible, ≈ 1-8 s (estimated field)
Operator action required	Crew dispatch and switching	Log review / supervision

6.6 Engineering Contributions

This study makes several engineering contributions; all grounded in the results demonstrated in Chapters 4 and 5 and none of which claims a field-ready protection product:

- a reproducible MATLAB/Simulink model of a representative 33/11 kV five-bus mining distribution feeder;
- a multi-bus RMS voltage and current feature framework (24 features) for feeder-state classification;
- combined fault-type and fault-location (zone) classification from bus measurements alone;
- a cost-sensitive Random Forest classifier tuned to penalise missed faults;
- a deterministic, auditable zone-to-breaker selective-isolation policy driven by the classifier;
- a closed detection-classification-isolation-restoration loop with topology-aware tie-switch restoration;

- an autonomous in-model FDIR controller that executes the full sequence within a single simulation;
- a quantitative comparison against conventional wide-area tripping; and
- explicit characterisation of the NER-limited and field-deployment limitations that bound the result.

These contributions are demonstrated under controlled simulation conditions; none constitutes a certified protection device.

6.7 Limitations

- **Near-metallic fault resistance only.** Fault resistances of 0.001-5.0 Ω do not represent the NER-limited 1-5 A SLG regime typical of properly earthed Copperbelt 11 kV systems, where the current rise that drives classification is greatly reduced.
- **Discrete parameter grid.** Training and test data share the same sweep grid; an off-grid interpolation test (Table 5.9) classified all 72 unseen parameter combinations correctly, though still within the near-bolted range tested.
- **Ideal sensors and communications.** RMS measurement is noise-free and instantaneous; a measurement-noise test (Table 5.8, Figure 5.14) showed accuracy remains at or above 95% (worst case) up to 5% per-reading error with no missed faults, but real metering also introduces bias, drift, saturation and communication latency not modelled here.
- **Unity-power-factor loads.** Loads are modelled at unity pf; a field feeder near 0.95 lagging would draw modestly higher current and alter the voltage profile.
- **Single-feeder radial topology.** Generalisation to other topologies requires separate validation.
- **No hardware validation.** The study is entirely simulation-based; hardware-in-the-loop testing is required before any deployment recommendation.
- **Idealised communications and no cyber-security modelling.** Measurements are assumed instantaneous and losslessly communicated; communication latency, communication loss and cyber-security threats present in a real SCADA-connected deployment are outside the simulation scope and are treated as deployment risks in Section 4.7 (Table 4.12, entries R5 and R6).
- **Ideal breaker and tie-switch actuation.** Breakers and the normally-open tie are modelled as ideal switches; mechanical operating delay, contact-resistance variation and breaker-failure/status-feedback behaviour are not represented, so the practical restoration timing in Section 4.7 is an estimate.
- **Missing or corrupted measurements not modelled.** The classifier always receives a complete 24-element feature vector; sensor dropout, saturation or missing channels, which a field system would need to detect and reject, are not simulated.

6.8 Recommendations for Future Work

- **NER-limited fault modelling (highest priority).** Add a neutral earthing resistor and regenerate the dataset with realistic 1-5 A SLG currents; sequence-component features (I0, I1, I2) will likely be needed to recover accuracy.

- **Extended robustness testing.** The initial off-grid interpolation and measurement-noise tests (Section 5.7) should be extended to the NER-limited regime, to correlated and biased sensor errors rather than independent Gaussian noise, and to a wider off-grid parameter range.
- **Hardware-in-the-loop validation.** Implement the classifier on an RTDS or OPAL-RT platform connected to real relay and breaker hardware.
- **Multi-fault and evolving scenarios.** Extend the FDIR logic to evolving faults, concurrent faults, and intermittent arcing.
- **Network generalisation.** Investigate transfer learning for diverse Copperbelt feeder topologies.
- **Field pilot study.** Begin with a monitoring-only deployment to gather real fault-event data before enabling automated actuation.

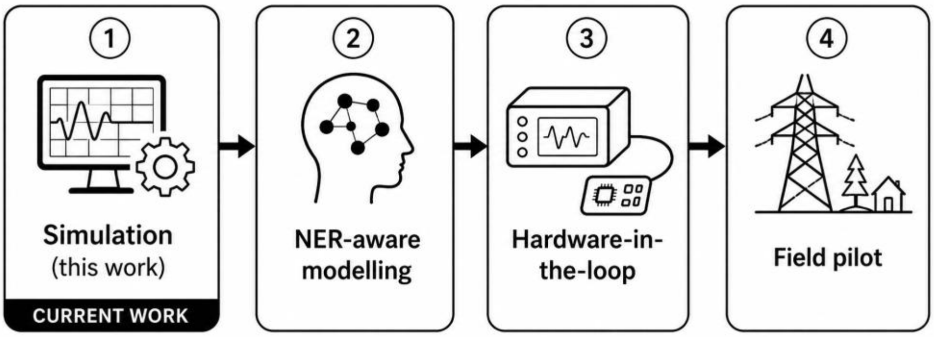

Figure 6.3: Future-work roadmap

Figure 6.3 organises the recommendations into a staged path, distinguishing the further simulation work that can be undertaken immediately from the steps that require laboratory hardware and field access before operational use. For industry, the practical guidance is to apply ML-based protection first in monitoring or advisory mode, to validate any classifier against site-specific earthing practice (especially NER-limited SLG faults) before enabling actuation, to retain conventional relay protection as the primary safety layer, and to collect real disturbance records for comparison before changing operational settings.

6.9 Concluding Remarks

This thesis has demonstrated that, under controlled simulation conditions, RMS voltage and current measurements drawn from multiple feeder buses can support machine-learning-assisted fault classification, selective isolation, and topology-limited service restoration on a representative mining feeder, with every switching decision driven by the classifier prediction. Its principal contribution is not a field-ready relay product but an integrated, reproducible engineering framework that links feeder modelling, fault-data generation, Random Forest classification, breaker selection, and conditional tie-switch restoration within a single study. This constitutes an appropriate first stage of investigation where high-voltage test infrastructure and real mine fault records are unavailable, and the work should be advanced through NER-aware modelling, interpolation testing, hardware-in-the-loop validation, and monitored field trials before any operational deployment is considered. In protection terms the scheme is best described as an FDIR automation layer that supervises and complements conventional relay protection rather than replacing it; the certified overcurrent protection remains the primary, fail-safe layer.

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Appendices

Appendix A Code, Data and Reproducibility

This appendix replaces lengthy printed MATLAB source listings with a concise reproducibility statement. The complete workflow, the Simulink model (mining_feeder_layer_FINAL_baseline.slx), the three master scripts, the labelled dataset (fault_dataset_v2.csv/.xlsx), the trained model (rf_model_v2.mat), the result summaries, and the final thesis figures are retained in the project archive. The workflow uses a fixed random seed (rng = 42) so that the dataset split and model training are repeatable within MATLAB R2024a with the Simscape Electrical Specialized Power Systems library. My public code repository is available at https://github.com/vicMenma/FYP_ML_Self_Healing_Mining_Feeder_Code. Table A.1 maps the repository layout, and the scripts are designed to run in the order shown in Table A.2.

Table A.1: Repository contents

Path	Purpose
`src/`	MATLAB scripts and the Simulink feeder models (mining_feeder_layer_FINAL_baseline.slx and the autonomous mining_feeder_layer_FDIR.slx).
`outputs/dataset/`	The 1000-sample labelled dataset (.csv, .xlsx, .mat): 24 features, 13 classes.
`outputs/model/`	The trained Random Forest (rf_model_v2.mat) with the confusion, out-of-bag, cross-validation and feature-importance data.
`outputs/waveforms/`	Twelve live-simulation fault-and-restoration captures (wave_{SLG,LL,3PH}_B{2-5}.mat).
`outputs/figures/`	The final thesis figures, organised by chapter, plus figure_manifest.csv.
`outputs/summaries/`	Block discovery, SLG grounding pre-flight, RF metrics, restoration results, the rule-based-baseline comparison, and the pipeline log.

Table A.2: MATLAB script execution order

Order	Script	Purpose
1	MASTER_A_PREFLIGHT_AND_DATASET.m	Discovers blocks, runs the SLG grounding pre-flight, applies the grounding fix, and generates the labelled dataset.
2	MASTER_B_TRAIN_AND_RESTORE.m	Trains the cost-sensitive Random Forest and runs the closed-loop restoration and waveform-capture scenarios.
3	MASTER_C_GENERATE_ALL_FIGURES.m	Regenerates the thesis figures from the stored outputs.
Analysis	ABLATION_COST_SENSITIVITY.m	Retrains a standard (no-cost)

		Random Forest on the existing dataset to isolate the effect of the cost matrix (research question 2). No re-simulation.
Analysis	NOISE_ROBUSTNESS.m	Evaluates classifier accuracy versus measurement noise on the existing held-out test set. No re-simulation.
Analysis	INTERPOLATION_TEST.m	Tests generalisation at off-grid fault resistances, load multipliers and onset times using the existing trained model.
Analysis	BASELINE_RULE_VS_RF.m	Compares the Random Forest with a max current-ratio zone rule and a single decision tree on the held-out test set (no re-simulation).
FDIR	BUILD_FDIR_CONTROLLER.m	Adds the autonomous in-model FDIR controller (detect -> classify -> isolate -> restore) to a copy of the feeder model, mining_feeder_layer_FDIR.slx.
FDIR	LIVE_FDIR_DEMO.m	Interactive command-line demonstration of the scheme for one chosen fault; restores the model state afterwards.
FDIR	getRF.m, classifyRF.m, reportFDIR.m, PATCH_FDIR_IDLE.m	Helper functions used by the in-model controller (load the forest, classify, report the result, set the idle display value).
Optional	RUN_ALL_PIPELINE.m	Runs the complete workflow (dataset → training/restoration → figures → analysis) with automated stage control and a smoke test.

The development environment used to produce the reported results is summarised in Table A.3. It is the environment used to build and run the simulation and is distinct from the field hardware that an industrial deployment would require (Appendix C).

Table A.3: Development hardware and software environment.

Processor	Intel Core i5-6200U @ 2.30 GHz (2 cores, 4 threads)
Installed memory	8 GB
Operating system	Windows 11 Pro, 64-bit (build 26200)
MATLAB	R2024a (version 24.1.0.2537033)
Core toolboxes	Simulink; Simscape Electrical (Specialized Power Systems); Statistics and Machine Learning Toolbox; Stateflow
Reproducibility	Fixed random seed rng(42)

This archive is an academic record for a simulation-based Bachelor of Engineering thesis. It is not a certified protection-relay package and must not be used for direct control of live mining electrical infrastructure. Practical implementation would require relay-setting review, hardware-in-the-loop testing, protection-coordination studies, cybersecurity assessment, site acceptance testing, and approval by competent protection engineers.

Appendix B Project Development Plan and Gantt Chart

Table B.1 is the project Gantt chart. It traces the development of this project from the selection of the research topic in July 2025, following a site visit to Konkola Copper Mines, through to final thesis submission in August 2026. The plan behind it is held in the spreadsheet Project_Gantt_Chart.xlsx, retained in the project archive described in Appendix A, which records the start date, finish date, duration, predecessor and percentage completion of every activity and presents the same plan as a bar chart. Table B.1 sets it out month by month.

Table B.1: Project development Gantt chart (July 2025 - August 2026)

Activity	2025						2026							
	Jul	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
1 Conception and proposal														
1.1 Topic conception (after KCM site visit)	■	■	■											
1.2 FYP proposal development			■	■	■	■								
M1 Proposal submitted						◆								
1.3 Concept development and refinement				■	■	■	■							
1.4 Scope simplification and title finalisation							■	■						
M2 Title and scope finalised								◆						
2 Modelling, dataset and machine learning														
2.1 Feeder Simulink modelling									■	■	■	■	■	
2.2 Topology rebuild and dataset generation												■	■	
M3 Labelled dataset complete													◆	
2.3 Machine learning and robustness analysis												■	■	
2.4 Autonomous FDIR implementation												■	■	■
3 Writing and submission														
3.1 Thesis drafting											■	■	■	■
3.2 Corrections, documentation and submission												■	■	░
M4 Thesis submission														◇

Note: A shaded cell means the activity was active in that month. Dark shading is work completed and light shading is work still outstanding at the status date of 10 August 2026. Rows M1 to M4 are milestones, marked in the month in which they fall; a filled marker is a milestone reached and an open marker one not yet reached. The numbered phase rows group the activities beneath them. Exact start and finish dates, durations, predecessors and percentage completion are held in the plan spreadsheet described above.

The plan is organised as a three-level work breakdown. The numbered phase rows (1, 2 and 3) group the work; the decimal rows are the individual tasks, each with a predecessor recorded in the spreadsheet; and the rows numbered M1 to M4 are the milestones against which progress was judged. The calendar runs across the top of the table, so the chronology is read from left to right. The 2026 columns also show that thesis drafting and the corrections ran concurrently with the modelling and machine-learning work rather than after it, which is how the project was actually carried out.

The project evolved from an initial concept - trust-based self-healing for mining transmission networks using IoT and machine learning - through progressive simplification to the implemented scope: a simulation-based, machine-learning-assisted self-healing scheme for a single 33/11 kV mining distribution feeder. The core Simulink implementation began in March 2026 and was rebuilt to its final topology, with the full labelled dataset, the Random Forest classifier and the autonomous in-model FDIR controller, in July 2026; thesis drafting, supervisor corrections and documentation continued in parallel from May to August 2026.

Appendix C Indicative Field Deployment Requirements

Appendix A records the environment in which the reported results were produced. This appendix answers the separate question raised there and in Section 4.7: what a field implementation would require. It is indicative rather than prescriptive. No vendor selection, procurement specification, protection-coordination study or costing has been carried out, and none is claimed. Its purpose is to state in engineering terms the distance between the simulation reported in Chapters 4 and 5 and an installation on a live 33/11 kV mining feeder. No cost estimate is offered, because the equipment schedule below has not been fixed and site-specific outage-cost data were not available for this study; quantifying the economic case is identified as future work in Section 6.8.

C.1 Deployment Architecture

Figure C.1 shows the indicative architecture. The scheme is supervisory. The protection IEDs at B2-B5 retain their conventional IDMT overcurrent and earth-fault functions and operate independently of the classifier at all times, so a failure of the FDIR controller degrades the feeder to the protection it has today rather than leaving it unprotected. Measurements travel upward from the instrument transformers through the station bus to the controller, which assembles the twenty-four RMS features, classifies the feeder state, applies the deterministic zone-to-breaker policy of Section 3.5 and evaluates the restoration interlocks of Section 3.6. Commands travel back down the dashed path, which is physically inhibited except in supervised-control mode.

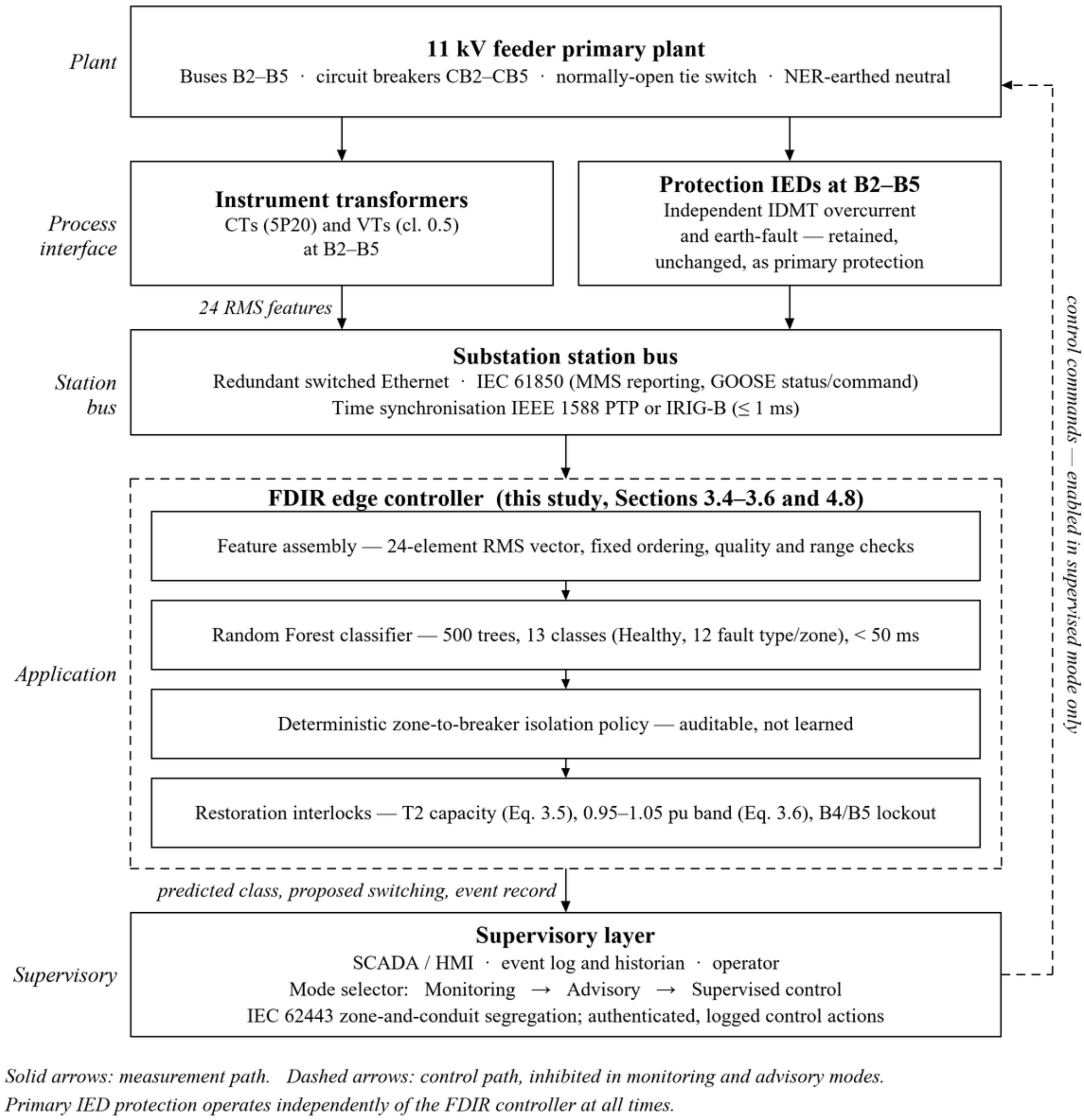

Figure C.1: Indicative field deployment architecture for the proposed FDIR scheme.

Two features of the architecture follow directly from the limitations recorded in Section 6.7. The feature-assembly stage carries explicit quality and range checks, because the simulation always presents a complete and noise-free feature vector whereas a field scheme must detect and reject missing or saturated channels. The classifier output is also kept separate from the switching decision, so that the learned component proposes a zone while the deterministic logic, which a protection engineer can review line by line, issues the commands.

C.2 Indicative Hardware Requirements

Table C.1 lists the plant and equipment a field implementation would require beyond the feeder's existing assets. Ratings are indicative and would be fixed by a site survey and a protection-coordination study.

Table C.1: Indicative field hardware requirements

Item	Indicative requirement	Driver, and relation to this study
Current transformers	One three-phase protection-class set (5P20) per monitored bus B2-B5, rated for feeder full-load current with adequate saturation margin.	Source of the twelve RMS current features. The accuracy class sets the measurement-error floor examined in Table 5.8.
Voltage transformers	Three-phase 11 kV/110 V units, metering class 0.5 or better, at B2-B5.	Source of the twelve RMS voltage features and of the 0.95-1.05 pu band check.
Protection IEDs	Numerical feeder relay at each of B2-B5, retaining independent IDMT overcurrent and earth-fault protection.	Primary protection is retained unchanged; the proposed function is supervisory (Section 4.7, Table 4.12 R1).
Neutral earthing resistor	As determined by the site earthing study; earth-fault current typically limited to 1-5 A.	Defines the regime the present dataset does not represent, and therefore the retraining requirement (Section 6.7, Table 4.12 R3).
FDIR controller	Substation-rated computer or rugged edge device: four cores, 8 GB RAM, no GPU; deterministic execution with a scan period of 20 ms or better.	Random Forest evaluation of 24 features was measured at under 50 ms (Section 4.7); the development machine of Table A.3 is already sufficient for the inference load.
Station communications	Redundant switched Ethernet station bus, fibre within the substation.	Carries measurements and commands within the 0.2-2 s allowance of Section 4.7 (Table 4.12 R5).
Time synchronisation	IEEE 1588 PTP or IRIG-B, 1 ms or better.	Aligns the per-bus RMS measurements into a single simultaneous feature vector; the simulation assumes simultaneity implicitly.
Circuit breakers	Existing feeder breakers with motorised mechanism, 52a/52b status feedback and clearing time of 100 ms or better.	Matches the 40-100 ms assumed in Section 4.7. Status feedback is required for the breaker-failure case the model does not represent (Section 6.7).
Tie switch	Motorised, remotely operable normally-open tie with interlocking and a voltage or synchronism check; operating time 1-5 s.	Executes the restoration action of Section 3.6, and dominates the practical restoration time estimated in Section 4.7.

Item	Indicative requirement	Driver, and relation to this study
Supervisory interface	SCADA or HMI with event log and historian.	Displays the predicted class in advisory mode and provides the audit trail required by Table 4.12 (R10).
Auxiliary supply	Substation DC supply with UPS-backed power for the controller and communications equipment.	The controller must remain available through the disturbance it is intended to act on.

Note: The equipment listed is additional to, or a modification of, the feeder assets already modelled in Chapter 4. Quantities and ratings are indicative; no vendor selection or procurement specification has been undertaken, and no cost is implied.

C.3 Indicative Software and Configuration Requirements

Table C.2 lists the corresponding software and configuration requirements. The distinction from Table A.3 is important: Table A.3 is the environment in which the classifier was developed, whereas Table C.2 describes what would run, and what would be controlled, in the substation.

Table C.2: Indicative field software and configuration requirements

Item	Indicative requirement	Notes
Classifier runtime	The trained forest exported to a deterministic standalone runtime, for example generated C code or an equivalent embedded inference library.	No interactive MATLAB session belongs in a substation; execution time must be bounded and repeatable.
Feature pipeline	Fixed 24-element ordering identical to training, one-cycle RMS window, with range, plausibility and completeness checks on every input vector.	The completeness check is absent from the present study and is a stated limitation (Section 6.7, Table 4.12 R4).
Decision logic	The zone-to-breaker policy of Section 3.5 implemented in the controller's reviewable logic, not learned.	Keeps the switching decision auditable by a protection engineer; the learned component proposes a zone, it does not issue commands.
Restoration interlocks	T2 capacity check (Equation 3.5), 0.95-1.05 pu voltage check (Equation 3.6), B4/B5 terminal-zone lockout, breaker-status confirmation and operation timeout.	The first three are implemented and tested in this study; the last two are additional field requirements.

Item	Indicative requirement	Notes
Mode control	Selectable monitoring, advisory and supervised-control modes, with control outputs physically inhibited in the first two.	Implements the staged rollout of Section C.4; inhibition must be physical, not only configured.
Communication protocol	IEC 61850, using MMS for reporting and GOOSE for status and command, or DNP3 where an existing SCADA dictates.	Recorded in Table 4.13 as outside the present scope.
Security	IEC 62443 zone-and-conduit segregation, authenticated and logged control actions, and no remote path for model update.	Addresses Table 4.12 (R6); the absence of any cyber-security modelling is a stated limitation (Section 6.7).
Model lifecycle	Version-controlled model artefact recording the training dataset, random seed and file hash; revalidation required after any retraining.	Extends the reproducibility practice of Appendix A into operation (Table 4.12 R9).
Event logging	Time-stamped record of the feature vector, predicted class, issued commands and resulting outcome for every event.	Supplies the site data required by Stage 1 and the audit trail for periodic model review.
Offline environment	The development environment of Table A.3 retained off-site for retraining, revalidation and figure regeneration.	Retraining must not be possible from the substation.

Note: Items whose notes refer to Section 6.7 correspond to functions that the present simulation does not implement; they are requirements on a deployment, not descriptions of work completed here.

C.4 Staged Commissioning

The staged rollout referred to in Section 4.7 is set out below. The sequence is deliberately conservative: no automatic switching is enabled until the classifier has been retrained on site data and tested against hardware.

- **Stage 1 - Monitoring.** The controller reads measurements and logs a prediction for every disturbance, with all outputs disabled. The purpose is to accumulate the site-specific event record, including the NER-limited earth faults that the present dataset does not contain (Section 6.7).
- **Stage 2 - Retraining and revalidation.** The classifier is retrained on the site data and re-tested by the procedure used in Sections 5.4 and 5.7: stratified held-out evaluation, five-fold cross-validation, a measurement-noise sweep and an off-grid generalisation test.

- **Stage 3 - Advisory.** The predicted class and the proposed switching are presented to the operator, who executes them manually. Agreement between operator and scheme is recorded and reviewed before any further step.
- **Stage 4 - Hardware-in-the-loop.** The controller and relay hardware are tested against a real-time simulator, as recommended in Section 6.8, including the breaker-failure, communication-loss and sensor-loss cases that the present model does not represent.
- **Stage 5 - Supervised control.** Automatic isolation and restoration are enabled with primary protection unchanged, subject to a protection-coordination study, site acceptance testing and approval by a competent protection engineer.

None of these five stages has been carried out. What the present study establishes is the stage before them: that the scheme is internally coherent, that its decisions are correct within the modelled conditions, and that its computational cost is negligible beside the plant it would command. This appendix exists so that the distance from that result to an operational installation is stated plainly rather than left to the reader's inference.